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Coffee Diseases

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4.1

Introduction

This chapter will only discuss coffee diseases due to fungal and bacterial infections. Diseases induced by nematodes and viruses will be discussed in the following chapters in Part II.

4.1.1

General Comments

Due to their different reactions to specific environmental conditions, the impact of the vectors of major coffee diseases varies greatly according to the environment. A typical example of this phenomenon is the case of Orange Leaf Rust (*Hemileia vastatrix*), a disease which is far more serious in hot climates and, therefore, less active at higher altitudes. Another typical case is that of the Coffee Berry Disease (CBD) found mainly in plantations at over 1500 m in Equatorial regions. Here, lower temperatures are not only favorable for the development of the pathogen, they also slow down the ripening process of the fruit which means that the berries are vulnerable for a longer period [35]. It is, therefore, important to bear in mind that the fact that a given pathogen is prevalent

does not necessarily mean that it will cause a serious disease with major economic consequences.

In certain cases, even if the plant shows symptoms of the disease, these only reveal a physiological disorder which, by weakening the plant, makes it more receptive to the disease itself. This is typically the case for “die-back”, due to *Colletotrichum* sp., or for Cercospora Blotch, due to *Cercospora coffeicola*. In both cases, as the parasites have only a secondary influence, the problem can be better solved by improved agricultural care.

Recent research, conducted in Honduras, could lead us to conclude that a better knowledge of plant physiology and the influence of certain environmental factors on the plant will enable us, in due course, to determine how nutrition in the coffee plant can influence certain diseases. Such knowledge could then be included in the measures taken to prevent the disease. Improved plant nutrition or appropriate agricultural practices may not eradicate the disease, but could certainly reduce its impact and lessen the possibilities for it to develop. Knowledge of this can help to reduce the use of fungicides for diseases control.

In any case, determining the factors which favor the development of diseases

should be of great help when recommending measures to be taken for fighting diseases in specific environmental conditions and areas [4].

Some diseases only attack the plant where it has been accidentally injured by agricultural tools, especially when the cut is at the base of the trunk, which often happens when cleaning weeds with a machete. Other injuries caused by insects or animals, as well as cultural practices like pruning or trimming, can also favor pathogen development. Obviously, care must be taken to avoid damaging the trees; however, where this is unavoidable, wounds should be treated with an antiseptic healing paste. This simple, but efficient, preventive measure should be a regular practice on all plantations as it also offers real protection against serious diseases like *Fusarium* wilt and canker.

Certain diseases like Coffee Wilt Disease due to *Fusarium xylarioides* which were a major preoccupation at a given time, seem to have disappeared over a period of 20 years, but have now resurfaced in certain African countries like the Democratic Republic of Congo and Uganda. From the 1930s to the 1960s, this disease, which attacked *Coffea excelsa* as well as *C. canephora*, was the main subject of a number of surveys and research in West and Central Africa. Then, for no discernible reason, it almost disappeared. It is generally thought that this can be explained by the introduction of a new, naturally resistant, variety of *C. canephora*, i.e. Robusta, coupled with the associated rejuvenation of plantations as well as improved agricultural practices including appropriate fertilization and better sanitary control. The momentary disappearance of this threat should not, however, lead to the belief that the disease is definitely eradicated. In fact, it reappeared in 1995 in the north-east region of the Democratic Repub-

lic of Congo and in 1993 in Uganda, where it has become a lethal threat to Robusta. At present, it has reached the stage of an epidemic disease and, so far, it has not been possible to determine whether this is due to mutations in the fungus or to a lack of care in agricultural practices and sanitary control, or, more likely, a combination of both factors. Whatever the reasons, the situation goes to show that vigilant supervision should never let up. In the south of Ethiopia, Coffee Wilt Disease on *C. arabica* now has an impact on the economy of the region.

Certain afflictions have been reported which cause the overall death of plantations for no identifiable reason. Research has not yet been able to pinpoint a common denominator for these disasters. This is the case for “sudden death” in Angola or the “mal de Viñas” in Guatemala.

4.1.2

Chemical Control

4.1.2.1 Pesticides

For a very long time, the available pesticides for disease control were limited to products based on copper sulfates, notably the famous and widely used “Bordeaux Mixture”, which is still frequently considered to be the reference fungicide. At present, however, the specialized chemical industry offers a large spectrum of different products which are not only preventive, like the copper sulfate compounds, but which, given their systemic or penetrating properties, also heal and act as an anti-sporulant. The fact that the systemics penetrate the tissues of the plant not only reduces the risk of them being washed away by rains, but also, if they are not degraded too rapidly, allows for enduring long-term action. An additional plus for these fungicides is the fact that they pollute

the soil far less as they deposit no copper, lead or other metallic residues.

This means that today we are well equipped to efficiently control the major leaf and branch diseases; however, pesticides are still not much help when it comes to fighting soil-borne diseases.

In spite of their many drawbacks, however, copper compounds are still popular and widely used, sometimes alternating with more recently developed products, for fighting leaf diseases. They may lack in penetrating action and require frequent, repetitive application due to rain, but they remain cheaper and, more importantly, they are less selective. Copper compounds act globally by coagulating the cytoplasm of the fungus, whereas more selective products have, in certain cases, generated new, resistant breeds of pathogens. The use of copper compounds alternating with new products has proved to be efficient [29, 49]. Copper compounds allow the new products to prolong their action and prevent the development of new mutant pathogens. The combination of the two has, over recent years, reduced the number of applications by about 30 %, or even 50 % in certain cases.

4.1.2.2 Productivity

In spite of the introduction of new pesticides with penetrating and systemic actions requiring fewer applications, chemical treatment remains exacting and expensive. The number of applications required involves labor costs, and the purchase and maintenance of special equipment, over and above the actual cost of the products. This, inevitably, increases productions costs and, frequently, entails the need for available labor at a time when work has to be done elsewhere on the farm. This is especially true for small holders.

Thus, the benefits of chemical control will be weighed against costs and productivity. Lack of treatment at a crucial time, too long an interval between treatments or a reduced concentration of the product in the mixture are essential factors which may well counteract all benefits.

Even if the cost of correct treatment is higher than the cost of defective treatment, the end result will be much better and profits will increase accordingly.

4.1.2.3 Epidemiology

Although the choice of the pesticide to be used is important, it is even more important to be well versed in basic epidemiology. It is necessary to know when to intervene, particularly when the first treatment should be made, and the sequence and the number of applications required. This can only be ascertained by careful observation and comparison of results carried out over a number of successive campaigns, taking into account the annual evolution of the disease under specific growing conditions. As all diseases caused by parasites develop according to climatic conditions and the vulnerability of the plants themselves, it is essential to determine the exact state of the plant combined with the climatic factors (rainfall and temperature) which spark off the development of the disease and the time it takes to spread.

For chemical treatment to be efficient, three key factors need to be respected:

- *When to begin.* Treatment must begin just when the infection sets in and at the time when climatic factors are most favorable to its spreading. This generally means that the effects of the disease are not yet clearly visible, but action is necessary, particularly when using preventive chemical products like copper compounds.

- *How long treatment is required.* Treatment must be sustained for the entire period during which the disease is likely to develop and spread. Treatment can be stopped just before the infection period peaks – a time which generally corresponds to a slow-down in plant growth and, occasionally, the end of the vulnerable period. If treatment has been sustained during the critical stages, the effects of the disease are reduced, and it is time to strike a balance between the costs of later treatments and potential benefits. One needs to judge whether the disease can still have an impact on either the year's production or that of the following year.
- *How often treatment is required.* Treatments need to be made sequentially in such a way that total protection of the plant is ensured during the critical period, this depends on the type and remanence of the pesticide used as well as on climatic conditions, particularly rainfall. For optimal results, the sequence of the treatments should be geared to the development of the disease and the specific rainfall conditions of the year, whereas, only too often, treatments are applied according to an empirically accepted pattern.

It is therefore of paramount importance that epidemiology be recognized as an applied science providing the necessary guidelines to ensure that chemical treatment is economically viable, efficient and cost-productive.

4.1.2.4 Treatment Methods

Application methods also need to be carefully chosen taking into account the areas which require treatment, the topography and the layout of the plantation. The equip-

ment can be manual or motorized, it can be carried or drawn, in large or small tanks, but in all cases it must be adapted to the layout and the needs of the plantation. The width between lines is a decisive factor as, for example, in larger coffee plantations, narrow-gauge tractors, similar to those used in vineyards, are quite suitable even for plantations where the lines are relatively close.

Another aspect that needs to be taken into account is the fact that fighting diseases requires high-volume spraying in order to cover, as far as possible, all the vulnerable parts of the plant. In this way, disease control differs considerably from pest control where very low-volume spraying or even fumigation is used. High-volume spraying, on the other hand, entails using large quantities of water. Even if part of the product sprayed, diluted by rains, frequently filters through to the lower leaves when dripping off the upper leaves, this cannot be relied upon to ensure total coverage. Therefore, a liberal use of spray mixture is necessary and, consequently, expensive. A number of experiments, particularly for fighting CBD, have proved this fact beyond doubt. Recently, however, good results have been reported on certain methods of low volume spraying used in fighting fungi, but this still needs to be checked out and confirmed.

Further experiments using mineral oils in conjunction with fungicides have been carried out in Cameroon as this type of "plantation oil" is used successfully in banana plantations for fighting or "Sigatoka disease" i.e. cercosporiosis. Treating banana plantations with "plantation oil" is carried out by spraying the mixture in a fine, regular spray from airplanes and, as the banana leaves which need covering are all at the top of the tree, this method is suitable. In coffee plantations, however,

the quantities of the mixture required to penetrate the thick foliage down to the lower leaves, which also need protection, are such that experiments resulted in burning the young leaves and the tips of the upper branches.

The need for significant quantities of water for high-volume spraying may create a problem since sufficient water storage is not always available on the spot when required. It is therefore recommended to build “impluviums” or water tanks on site, inside the plantation itself. For larger estates, permanent tanks or water towers serve the purpose; in smaller plantations, more primitive water collectors like a simple pit lined with a sheet of plastic and topped with corrugated metal plates set at an angle on each side to funnel the water are quite suitable.



Picture 4.1 Primitive rainwater collector “impluvium” on a small plantation.
From: J. N. Wintgens.

Another, very similar and simple solution to the problem would be a 200 l drum with similar corrugated metal or plastic plates to funnel the water into the container.

4.1.2.5 The Rates of Fungicides

When calculating the quantities of fungicide required, considerations like “kilograms of product (either active or commercial) per hectare” or “hectoliters of mixture per hectare” should be discarded. Specific calculations should be made taking into account factors like the actual size of the trees and their density of plantation. As these factors vary considerably from one plantation to another, commercial indications regarding the quantities required are of little use. Therefore, when counseling producers, one should think in terms of the concentration of the mixture (kilograms of product per hectoliter). The concentration should be calculated to ensure efficient maximum protection with a product input, allowing for the “dripping effect” from the upper to the lower leaves. In this way costs can be geared to real needs and the producer can learn to estimate quantities with greater precision.

4.1.2.6 Assessment of Damage related to the Efficiency of the Treatment

When setting up the most efficient method for disease fighting, making the right decisions is no easy task. On the one hand, it is necessary to assess the potential efficiency of the treatment and, on the other, to assess the potential damage risks if no treatment is implemented.

If, for example, the disease attacks the foliage, damaged leaves should be periodically counted in order to observe the evolution of the disease and to evaluate its spread [1, 43]. However, this type of assess-

ment does not always reveal the true impact of the disease. This aspect needs to be examined in terms of its effect on the potential annual yield. It is necessary to ascertain just how far the disease can be allowed to spread before chemical spraying becomes imperative and, if chemical spraying is implemented, whether the costs involved will be worth the investment. The only way to answer these questions is by comparing crop results on treated and untreated control plots in a given region.

The evaluation of damage due to disease at a given time is often complicated by the fact that significant damage can also be caused by physiological weaknesses in the plant and it is sometimes difficult to distinguish this from the true symptoms. Specific observation methods need to be implemented in order to come to the right conclusions and decide on what type of action needs to be taken.

4.1.3

Genetic Control

4.1.3.1 The Benefits of Genetic Engineering

Although the use of chemicals for fighting cryptogamic diseases is efficient, even very efficient if carried out correctly, this method still has to be continually repeated, year after year. For this reason, producers are beginning to examine the advantages of genetic engineering. It may well be more costly and take far longer to implement, but the end result is a durable protection and consequently represents a worthwhile investment.

4.1.3.2 Progress of Research on the Coffee Plant

The great diversity of planting material already existing within the various collections has been significantly increased over

recent decades by surveys throughout the African continent, and now represents an impressive gene bank available for use in the more traditional methods of hybridization and breeding.

Recent biotechnological progress may improve selection methods and create more resistant varieties. Studies in cellular and molecular biology have not only helped to identify and select quality genitors to be used for breeding, but have also brought about a more precise identification of pathogens, and a better knowledge of their geographical distribution and their genetic variability. Research can, therefore, be focused with greater precision and is now developing methods for using gene transfers to fight certain types of insects. The possibility of using similar strategies for fighting disease seems likely in the near future. *In vitro* propagation techniques, the use of micro-cuttings and somatic embryogenesis are techniques which now enable the early discernment of resistant genotypes issued from hybridization. These techniques bypass the need to wait for a number of generations to observe the fixed characteristic. When compared to traditional horticultural methods, they are, therefore, great time-savers in the reproduction process of the genotype.

Unfortunately, however, as reported in Section II.6, resistant characteristics do not necessarily become permanent acquisitions. The use of resistant varieties should be carefully monitored taking into account the fact that pathogens can also modify their behavior to overcome the plant's resistance and even amplify their virulence. Strategies exist to compensate for shortfalls in resistance and they must be implemented without fail.

4.1.4

Biological and Agronomic Control

Apart from the use of pesticides and genetically acquired resistance, it is also worthwhile to examine biological and agronomic methods of disease control.

4.1.4.1 **Prophylactic Measures**

Among the simplest management methods, healthy growing conditions are always beneficial in fighting disease. The sick parts of the plant are the areas where pathogens develop and breed so that, wherever possible, it is better to remove and burn this material of the plant in order to avoid further contamination. This was common practice in the old days when little was known about more sophisticated fighting methods and, even now, this practice should not be neglected. “Removing and burning” should be systematically carried out between production cycles as this method delays the inception of infection and, consequently, limits the number of reproduction cycles for the pathogen. It is even possible to “remove and burn” during the cycle which would efficiently prevent the infection from developing into an epidemic. In certain cases, particularly for “Pink Disease” or “Thread Disease” or even for fighting CBD in Arabica, this method is highly recommended. It is also valid when preparing the soil for a new plantation as it eliminates vegetal debris which could be contaminated by root rot, for instance.

Although prophylactic measures alone will not solve all problems, they will certainly reduce the impact of the disease and diminish the importance of the attacks as well as help to improve the efficiency of chemical treatment.

Prophylactic management methods like removing and burning or protecting accidental injuries with an antiseptic healing paste do not require undue effort, and should always be included in general maintenance practices.

4.1.4.2 **Agricultural Practices**

Apart from prophylactic management methods, cultivation practices should provide the most favorable environmental conditions for the crop. Here, it is important to distinguish between the different types of infection:

- In cases where the presence of pathogens reveals a physiological deficiency like, for example, “die-back”, it is obviously necessary to improve conditions by an adapted input of nutrients, suitable drainage, appropriate shade, etc.
- In cases where environmental conditions like excessive humidity or too much shade may not be detrimental to the coffee plant, but are conducive to the development of pathogens, pruning, trimming and suitable drainage should be implemented.

One must be aware, however, that these methods need to be practiced with caution as they could be more harmful to the plantation than the pathogen itself. In the case of Coffee Leaf Rust, for example, pruning will increase the effect of the sun on the plant and thus hinder the development of the pathogen, but, at the same time, it will increase productivity. This means that the plant will need an extra input of nutrients, particularly nitrogen, to compensate for its increased production. Furthermore, it has been proved that a higher production of fruit provides more scope to the spread of berry diseases like CBD. It is wise, therefore, to weigh the advantages against the

disadvantages when implementing certain cultivation practices to fight coffee diseases.

The above comments go to show that present-day agricultural practices are somewhat limited when it comes to fighting coffee diseases. In future, however, research may well improve our knowledge of specific physiological conditions of the complex host-parasite interaction where corrective and preventive cultivation methods could be implemented to efficiently control it. For the time being, however, we can only recommend that the cultivation of coffee plants takes place in ideal environmental conditions for the plant itself. This will avoid a number of setbacks and ensure a profitable crop to cover the costs of fighting disease.

4.1.4.3 Biological Control

Biological disease control is already used to destroy certain insects which attack plantations and, similarly, the idea of using natural enemies to destroy coffee plant pathogens has been explored for quite some time. Hyperparasites of the *Hemileia* sp., e.g. especially *Verticillium hemileiae* = *V. lecanii*, frequently accompany *H. vastatrix* whose spores it destroys. This also applies to *Paranectria* spp. which attacks the spores of *H. coffeicola*. To date, however, studies and research into these possibilities have not yet come to any practical conclusions which could be implemented.

For the control of soil born pathogens, the use of certain types of *Trichoderma*, reputed antagonists for a number of species, is being seriously considered. *T. harzianum*, which generates antibiotics that hinder the development of filamentous fungi and hydrolytics which destroy the mycelium of pathogens, could efficiently counteract Basidiomycetes (*Armillaria* sp.), As-

comycetes (*Rosellinia* sp.) and various other fungi of different groups like *Pythium*. Similarly, the highly competitive strain of a non-pathogen, *Fusarium oxysporum*, could be used to prevent the development of other microorganisms in the soil and also because, after inoculation, it produces phytoalexins in the coffee plant which generate a certain resistance. Biological control of soil pathogens with natural enemies can thus be viewed as a hopeful development for the near future, which would be a real breakthrough as soil pathogens are the most difficult to fight.

Some research scientists also hope to use strains of these fungi to fight foliage diseases as well, but here research is still at the hypothetical stage.

4.1.4.4 Escape Strategies

A typical example of “escape strategies” can be seen in the method used to reduce the propagation of CBD. This strategy is carried out by using early irrigation to bring forward berry ripening. The crucially vulnerable stage when the cherries develop is shifted to a time when natural climatic conditions are not favorable for the pathogen to become active, thus reducing infection [35]. This is, of course, a unique case; however, research could develop the idea and possibly discover similar escape strategies to be used in other cases.

4.1.5

Eradication Measures

When a new disease starts up in a given region, the first reaction is often to attempt to eradicate it immediately by destroying the first focus of infection. The only condition for successful eradication in this case is to be sure of catching the disease at its onset, before the pathogen has time to de-

velop and spread. Once it has started to spread, eradication becomes illusory. In the early stages, eradication does not only mean the destruction of diseased plants, but also the elimination of neighboring plants which may appear to be healthy but could already be carriers. To be certain of having destroyed the disease at its onset, a careful observation, on a daily basis, of the plants on all the plantations is necessary over a long period. This is a major task which, initially, could well proscribe the practice altogether.

Orange Leaf Rust (*H. vastatrix* Berk. and Br.) is a typical example of the need for vigilance when implementing escape measures.

This disease existed worldwide with the exception of America until 1970; however, the threat was greatly feared by all groups involved in coffee production on this continent and rightly so, at least to a certain extent. As all the plantations had originated from the same material, imported to the Caribbean and Guyana in the 18th century, and this source was known to be receptive to the disease, it was feared that once the pathogen reached the continent, the entire coffee cultivation would be condemned. Nevertheless, as it turned out, the fear was somewhat over stressed as experience had already shown that the great diversity of environmental conditions in the tropical areas of America would lead to an equally varied development of the disease, running from a total disaster to a relatively minor onslaught. Inevitably, when the disease first appeared in the state of Bahia in Brazil in 1970, there was a reaction of panic. Initially, it was decreed that all the coffee trees in the state should be destroyed and a safety belt stretching over 385 km between Rio de Janeiro and Bello Horizonte was set up to prevent the spread of the disease. Fortunately, things calmed down fairly quickly

when it was noted that there was no way the disease could be controlled simply by eradication as it had been caught too late and had already contaminated over 500 000 km², i.e. an area which was far too extensive to hope to control the spread by eradication [34].

In Nicaragua, Orange Leaf Rust was first detected in 1976 and eradication was immediately planned. In 1977, diseased coffee plants and those within a radius of 20–30 m were destroyed with chemical brush-killers or burnt. Plants on the fringe of the condemned zones were sprayed with a fungicide and the canopy of trees further away was treated preventively [47]. Although official reports claimed that the campaign was a success as the number of diseased trees had significantly decreased, the operation had to be repeated every year as new disease foci appeared. At the end of the campaign, as Orange Leaf Rust only persisted in one isolated zone, it was decided to destroy every plantation in the contaminated area (7000 ha) and to reinstall them on the basis of intensive cultivation methods with dwarf plants of the Caturra variety which, it was thought, could not be infected as coffee trees in the entire zone had been destroyed. In actual fact, one or two plantations escaped destruction so the costly operation was unsuccessful and Orange Leaf Rust had come to stay. The final result of this stubborn undertaking had done far more harm than the disease could possibly have done over many years. Potential productivity had been ruined in the entire area for a number of years and a great deal of money had been spent to create new plantations. A 55 ha nursery was set up which used 30 tons of seed and employed 4000 workers in an area which, although distant from the infected zone, turned out to be unsuitable for coffee growing. One way and another,

the operation turned out to be a total and costly failure.

It was, actually, the success of the first eradication campaign in Papua New Guinea [41] which led to the conviction that an eradication campaign could also be successful in the tropical zones of America. Conditions, however, were radically different. The insular isolation of Papua New Guinea and the scarcity of coffee plantations at the time were the major keys to the success of the first campaign there, whereas the entire American continent, with very extensive areas of coffee cultivation could not hope to meet with similar success. It should also be noted that when the disease re-appeared in Papua New Guinea in 1986, eradication was not even attempted as, by that time, coffee cultivation had developed to an extent where eradication was no longer justified.

The lesson to be drawn from these various examples is that, as a general rule, eradication campaigns are counter-productive and costly. It is wiser to learn to “live with the disease” and to use chemical or other methods available to keep it under control. It should also be remembered that the presence of any given pathogen does not necessarily lead to total disaster, particularly as environmental conditions frequently limit its impact.

4.1.6

International Exchange

To round off this introduction, it should be recorded that certain serious diseases, e.g. CBD, are still contained on one continent. This means that necessary precautions should be taken when transferring plants. This does not only apply to inter-continental transfers, but also to transfers from one country to another on the same continent, particularly as, due to differences in the

impact of local strains of the pathogen, the same disease has different levels of virulence in each area.

4.2

Cryptogamic Diseases

A great number of various types of fungus can be found on the coffee plant but, as many of them are insignificant, only six or even serious cryptogamic diseases need to be taken into account from an economic point of view. Others may also have their importance under specific climatic or agromomic conditions.

4.2.1

Nursery Diseases

4.2.1.1 Damping Off

French	Fonte des semis, pourriture du collet
Spanish	Volcamiento, mal del talluelo, Podredumbre del cuello, rhizoctoniasis
Portuguese	Tombamento, Estiolamento, Rhizoctoniose das mudas
German	Keimpflanzenkrankheit
Dutch	Kieimplantenziekte

This is the most frequent disease and the coffee plant is susceptible to attacks at a very early age. It occurs in seed beds and in nurseries alike. It is a type of rot brought on by *Rhizoctonia* – *Rhizoctonia bataticola* (Taub.) Butler, which is very widespread, or by *Rhizoctonia solani* Hühn, which is more often reported in the Americas. It is active in most tropical countries and attacks all species of coffee as well as tea, cocoa, rubber plants, etc.

In seed beds *Rhizoctonia* sp. Attack the collar, rotting rings around the bark of the tiny stem, thus preventing the sap



Picture 4.2 “Damping off” due to Rhizoctoniasis – observe the damage to the collar of the seedlings. From: Nestlé.

from circulating correctly and causing the seedling to wither away. The disease develops in round spots which spread fast in humid soil and can destroy entire seed beds in only a few weeks. In nurseries, the new young roots are attacked first. From there, the main roots are attacked, and, again, the plant dries up and withers.

The most appropriate preventive measure is to disinfect the soil before sowing or transplanting and the best traditional method for disinfecting is to use a solution of formalin at a concentration of 2–3 % sprinkled over the soil with a watering can. Other products like cryptonol are available now, but should only be used in strict compliance with the manufacturer’s instructions. In Central America, methyl bromide is used frequently with the added advantage of also acting on small seeds, other pathogens and the nematodes which are particularly aggressive in this area. Due to its toxicity, however, this product is likely to be forbidden in a near future, but it can be replaced by Basamid (Dazomet) which is similar but less dangerous and easier to apply.

Nurseries should also be carefully drained to avoid stagnant water and excess humidity.

Whenever the disease actually appears, the seedlings should be burnt and the

soil, as well as the other seedlings in the vicinity, should be sprayed with an appropriate fungicide.

4.2.1.2 Seedling Blight

Causal organism

Fusarium stilboides Wollenw.

Occurrence/distribution

The disease is common in germination beds and nurseries. It is found in most countries where coffee is grown.

Symptoms/disease development

On infected seedlings, the cotyledons often fail to unfold, the stem has necrotic lesions and the seedling wilts. In some cases, necrotic lesions appear on the stalk below the unfolded cotyledons. These lesions enlarge and eventually result in the death of the seedling.



Picture 4.3 “Seedling Blight” – observe the necrotic lesions on the stalk.

The fungus is seed-borne, and can also be spread by insects and rain. Infection is severe during periods of high and frequent rainfall. The optimum temperature for infection is around 25 °C.

Control

The control methods used are hygiene and chemical treatment. Disease-free seeds should be used. Seeds may be treated with Benomyl (1 g of Benomyl 50 % WP/kg seeds) before sowing.

4.2.2

Root Diseases

4.2.2.1 Root Rot Diseases

French	Pourridi�s
Spanish	Podredumbre de las ra�zes
Portuguese	Pedridas da ra�z
German	Keimkrankheiten des W�rzelsystems und der Stammbasis

Causes and symptoms

Major root afflictions, in coffee trees and in most other ligneous species, can be ascribed to very polyphageous fungi, chiefly Basidiomycetes and certain Ascomycetes, more popularly known as “root rotters”.

Root rotters are active in a number of very varied conditions and their polyphageous nature indicates that environmental conditions like vegetation (forests) and cultivation practices (the presence of shade trees) are determining factors for their development.

Injuries to roots brought about either by insects or nematodes or, more importantly, by mechanical tools used for cultivation open the way for the fungi to penetrate the roots and to spread from plant to plant. Generating from specific localized sites, the fungi progress via rhizomorphous mycelium, web-like growths that stretch from root to root. The natural evolution of the mycelium is generally a relatively slow process; however, where mechanical tools spread fragments of mycelium, evolution accelerates and the fungi propagate rapidly through the rows.

This phenomenon was noticed and reported in a high-altitude plantation of Arabica in Cameroon (1650 m) which was afflicted by *Clitocybe elegans*. The first signs of the disease were pinpointed in 1957 and after 9 years only 9 % of the trees were diseased. In the following 4 years (1966–1970), however, with more extensive use of mechanical tools, a further 10 % of the trees was infected. This represents a fairly typical exponential curve.

Root rot is fairly easy to detect as the leaves wither visibly and the branches decay into dry rot. The plant weakens and finally dies. These diseases are serious as they directly affect the crop and the productivity of the plantation often involving a significant loss of investment capital. The death of the afflicted trees can take place within a few months only, but depends on the hardness of the wood. Diseased trees may live on for a number of years, although the speed at which the disease brings about the death of the tree is directly related to the length of time it has taken to mature. Thus, an Arabica grown at high altitude will resist longer as it has taken longer to develop and its wood is denser, whereas the same species of Arabica grown at a lower altitude will have matured faster and its wood will, therefore, be more vulnerable to rapid decay by root rot.

The causes can be precisely identified by examining the roots and the base of the tree trunks. The more common pathogens can be described as follows:

- Two Ascomycetes, *Rosellinia bunodes* (Berk. and Br.) and *Rosellinia pepo* Pat., generally develop best where the collar of the tree has been injured. Known as black root rot, (French = Pourrid  noir, Spanish = Podredumbre negra, Hongos negros de las ra zes, Portuguese = Mal de quatro a os, Podrid o negra, roseli-



Picture 4.4 *Rosellinia* sp. – black root rot.
From: G. Castillo.

niose) these fungi generate black rot on the bark and roots where a black mycelium develops groups of round, blue-black spots. This disease is particularly prevalent in El Salvador and although it is relatively unimportant in neighboring countries, who is to say that it could not prevail elsewhere under similar environmental conditions?

- *Phellinus lamoensis* (Murr.) Heim, causes brown rot of the root tissues (French = Pourriture brune des racines, Spanish = Mal pardo de las raíces, Portuguese = Podridão parda). Its mycelium mixes with the soil to form scabs on the outer tissues of the root.
- *Leptoporus (ex-Fomes) lignosus* (Klot.) Heim ex-Pat., causes white root rot of all tissues. (French = Pourridié blanc, Spanish = Podredumbre blanca, Portu-

guese = Podridão branca) The whitish mycelium webs cling to the diseased tissues and along the roots generating small ledges, built by carpophores, at the base of the trunks. This type of rot is prevalent in Africa.

Three types of agaric fungi, often confused by non-specialists, *Armillaria mellea* (Vahl) Quel., or *Armillariella mellea* (Vahl) Pat., which is very widespread, *A. fuscipes* Petch. and *Clitocybe tabescens* (Pers. ex-Fr.) Bres or *Armillaria (Clitocybe) elegans* R. Heim, Bres. These fungi, generally known as agaric rot (French = Pourridiés agarics, Spanish = Podredumbres agaricos) generate an easily recognizable pathology called “split disease” where the diseased organs like roots, collars or stems, split lengthwise.



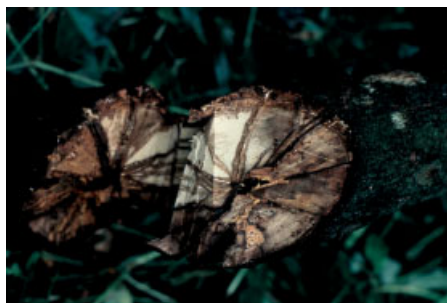
Picture 4.5 Agaric root rot (*Clitocybe elegans*):
splitting the collar and the trunk of a coffee tree.
From: G. Blaha – CIRAD-CP.



Picture 4.6 Agaric root rot (*Clitocybe elegans*): carpophores of the fungus emerging from the split of the collar and the trunk of the coffee tree. From: G. Blaha – CIRAD-CP.

“Root splitting”, “collar crack” or “stem splitting” comes from the mycelium generating spikes or xylostomes of a few millimeters in diameter with carpophores along the medullar rays of the core of the diseased organ, causing it to split.

C. elegans is prevalent worldwide. In the 1940s and 1950s it caused very serious problems in Madagascar where it attacked both Robusta and Kouillou coffee plants as well as the surrounding shade trees, particularly *Albizia lebbek*. Likewise, in Cameroon [10], where it attacked Arabica plantations, including the surrounding shade trees (*Albizia malacocarpa*, *Leucaena glauca*).



Picture 4.7 Agaric root rot (*Clitocybe elegans*): cross-section of a coffee tree trunk showing the xylotomes of the fungus developed along the medullar rays. From: R. A. Muller.

Root rot control: curative measures

Overall, one can do little more than limiting the spread of the disease. As the pathogens develop via their rhizomorphus network from root to root, and thus from plant to plant, attempts to restrict the spread from an initial site where the disease is festering can be made by digging a circular ditch 60 cm deep around the site to cut the web. When digging the ditch, the contaminated soil should be thrown towards the inside of the circle, not the outside. In practice, the diseased trees and their surrounding shade trees are pulled out.

Special care should be taken to remove as much of the woody roots as possible using a levered hoist if available. Trunks, twigs and roots must burnt on the spot. Before re-planting any coffee trees, grass, leguminous plants or any other crop impervious to fungi diseases should be sown over the dug up area to prevent erosion and leaching. Leaving the land fallow has a cleansing effect, but it is slow as illustrated by Tab. 4.1 compiled after an experiment carried out in Cameroon to follow the development of *C. elegans* on a high-altitude Arabica plantation [10]. This method consisted of placing small sticks previously

Table 4.1 Initial results of the “trap-stick” experiment

Period during which the land was left fallow (months)	Percentage of contaminated sticks at a depth of	
	40 cm	80 cm
2	41	59
12	34	35

sterilized and cut from wood of a species receptive to cryptogamic diseases like Hevea or Leucaena in the contaminated soil. The sticks (30–40 cm long and 3–4 cm in diameter) should be placed at varying intervals and depth. If the fungus is still present, they will be rapidly contaminated, hence their nickname: “trap-sticks”.

The experiment clearly indicates that “natural” cleansing of the land is a slow process and that it would be a wise precaution to avoid replanting coffee trees in contaminated areas for at least 3–4 years.

Chemical decontamination can, however, accelerate the process. The holes made when uprooting the diseased trees can be treated with Basamid or even methyl bromide, which is commonly used in Central America.

In-depth fumigation of contaminated soil with methyl bromide has also been carried out experimentally on high-altitude Arabica plantations in Cameroon and has shown positive results in the destruction of fragments of *C. elegans* mycelium which remain in the soil after the tree has been removed. Although the product does not seem to have a direct effect on the mycelium itself, it accelerates the generation of *Trichoderma*, a natural enemy, resistant to methyl bromide, which efficiently destroys the mycelium.

From an economic point of view, this method is only valid for small, localized

sites, but it can certainly be recommended when the disease has been pinpointed in its early stages.

From a practical point of view, the best way to implement the method is to loosen the soil with a hoe and then cover the hoed area with a plastic film. Once this is securely in place with the edges tucked into the soil to make sure it is airtight, insert a pressurized cartridge containing the methyl bromide gas (1400 g) under the sheet, open it and let the gas permeate the soil. This method is efficient on 9 m² (3 m × 3 m) areas centered on the hole where the tree was uprooted. The operation should be repeated once a week for 3 consecutive weeks. When the treatment is spread out over a period of 3 weeks, better results have been obtained than when a more concentrated dose is injected in one treatment only. Table 4.2 illustrates this point.

Even after treatment of a given site, contamination may still spread. This can be monitored by placing trap-sticks on the fringe of the treated area or even further outside the zone. Any residual contamination will be revealed by the white mycelium on the trap-sticks.

Table 4.2 Results after 2 months treatment with methyl bromide (obtained by the “trap-stick” method)

Quantities of methyl bromide used on 9 m ² (g)	Number of applications (with an interval of 1 week)	Percentage of contaminated trap-sticks at a depth of	
		40 cm	80 cm
1400	1	17.70	19.38
	3	1.02	22.45
2800	2	5.10	9.18
	3	0	4.08

Another method which can be used to limit the extension of root rot is to ring the bark of the trunks of healthy coffee trees in the immediate vicinity of the infected area. The plant can survive for a time on its reserves, but the root rot pathogens, which survive on starches and not on the lignin, starve. This practice entails sacrificing at least one row of healthy coffee trees around the infected area and would obviously be far too costly if many infected areas were scattered throughout the plantation.

In conclusion, there is no simple recipe for fighting root rot, but, possibly, future research should concentrate on encouraging the presence of natural enemies like *Trichoderma*, particularly *T. harzianum* or non-pathogenic strains like *Fusarium oxysporum*.

Once the plantation has been completed, control is still required to detect the development of any pocket of infection so as to eradicate it when still in its early stages. Residual stumps and/or dead wood and even certain shade trees may still expose the young plantation to contamination.

After destroying the stumps and dead wood, a further precaution would be to cultivate a leguminous crop like peanuts or beans, etc., for 3 years on the cleared land prior to planting the young coffee trees. This practice, although not generally very popular with producers, has the merit of cleansing the soil more thoroughly.

4.2.2.2 Mealybug Root Disease or Phtiriosis

French	Phtiriose
Spanish	Ftiriosis
Portuguese	Ftiriose
Dutch	Wortelluisschimmel

Known worldwide, this sickness is caused by different species of Mealybugs (*Pseudococcus* sp., *Formicoccus* sp., etc.) spread by

ants (*Paratrechina* sp., *Pheidole* sp. and other similar species). The bugs colonize the collar and the roots of the coffee trees. The conidial phase of *Bornetina coryum* develop on their sweet excretions and, by mingling with the surrounding soil, generate a brown mycelium sleeve-like coating which encircles the vital parts of the plant.

The afflicted coffee trees are weakened and chlorotic. In extreme cases, where the stifling effect of the external mycelium sleeves combined with the internal action of the bug on the sap are not brought under control soon enough, the plant suffocates and dies.

Here, obviously, preventive measures should target both the bug and the ants. The most effective method of control is to sprinkle the base of the coffee trees with a liquid solution of Aldrin or Dieldrin (3–5 g per tree).

4.2.3

Diseases of Aerial Organs

4.2.3.1 Coffee Wilt Disease

French	Tracheomycose
Spanish	Tracheomicosis
Portuguese	Traqueomicose, Fusariose

Causal organism

Fusarium xylarioïdes Steyaert, *Gibberella xylarioïdes* (Steyaert) Heim and Saccas, or *Carbuncularia xylarioïdes* (Heim and Saccas).

General comments

this disease was first detected in 1927 in Oubangui-Chari (now the Republic of Central Africa) and was initially thought to be caused by root rot. The true pathogen was only identified in 1939 by examining samples of dead coffee trees from the Democratic Republic of Congo, former Belgian

Congo. In the Central African Republic the disease spread and developed drastically over the following decade. Initially it chiefly affected Excelsa coffee trees, but by 1949 it also affected *C. neo-Arnoldiana* [46] and, to a lesser extent, *C. canephora* var. *robusta*. At the same time, an overall deterioration of coffee trees in the Ivory Coast was also attributed to this disease where it affected the two local species of *C. canephora*: the Kouillou Bandama and the Kouillou Touba as well as *C. abeokutae*, also called Indinié.

All in all, the disease had extremely serious effects in Central and West Africa from 1938 right up until the 1950s. In these areas, it practically wiped out the entire cultivation of *C. excelsa* and similar species like *C. abeokutae*. For *C. canephora*, the Kouillou variety was particularly vulnerable to the diseases and, whereas the Robusta variety showed greater resistance to the disease in West Africa [33], it seemed more vulnerable in Central Africa. This leads us to suppose that the pathogen has considerable variations in its virulence.

Researchers and producers alike concentrated their efforts on investigating this disease during the crucial period, but by the mid-1960s interest waned and the disease was practically forgotten. The chief explanation for the decline of the disease can be attributed to the fact that the initial vulnerable plantations which had been wiped out by the epidemic were replaced by the more resistant Robusta variety and also to the regeneration of the plantations, coupled with the development of improved methods of control. In Central Africa, however, Robusta is still more vulnerable to the disease than in West Africa. This can only mean that the disease is still latent and therefore vigilant supervision is still necessary as there is no way to predict how the pathogen will evolve.

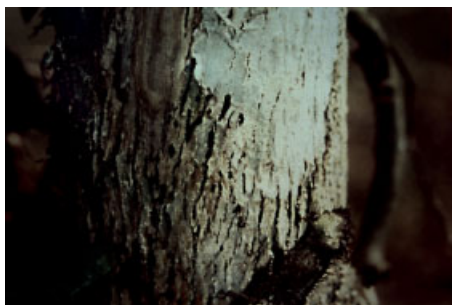
The disease was first reported in 1957 (Stewart) on *C. arabica* in Ethiopia and has spread very little to date. The disease re-surfaced in Democratic Republic of Congo in the middle of the 1980s and had a significant impact in the east and in the province of Kivu. The disease was reported in Uganda in 1993 and nearly all the coffee areas of this country were attacked. The reason why the disease resurfaced is unknown. Various hypotheses have been suggested like bad sanitary conditions in the plantations or a mutation of the pathogen. Detailed investigation will be required to explain this phenomenon.

Symptoms and biology

The leaves turn yellow, dry up and fall, although sometimes this only happens on part of the canopy (hemiplegia), branches die, and, finally, within a few months or even a few weeks, the entire tree withers and dies. These symptoms, however, can be classed as “first-degree” symptoms. Common to a number of afflictions and potentially due to a number of different causes, they cannot lead to a precise diagnostic of the disease.

“Second-degree” symptoms, more specific to the disease itself, therefore need to be examined. They can be found on the trunks, lodged in small cracks of the bark where tiny blue-black pellets, the perithecia of the pathogen, develop. One way of checking the disease is to examine the layer underneath the bark where is a long black strip bordered by healthy areas.

The decay of the coffee plant is due to colonization of the sap vessels by the mycelium, and the tyloses that block water and sap circulation. The infection can set in any time from the cotyledon stage to maturity. The death of the tree depends on the age, the vigor of the plant or the agro-environmental conditions and can occur



Picture 4.8 Tracheomycosis on the trunk of *C. canephora* var. *robusta*. From: R. A. Muller – CIRAD.



Picture 4.9 Tracheomycosis on *C. canephora* var. *robusta*. Observe the black, brownish or purple color of the section. From: GENAGRO.

within 9–18 months. Both micronidia and macronidia abound shortly before death and when the tree dies. The infection spreads via the micronidia, the macronidia and the ascospores because the perithecia continue to develop for a long time after the death of the tree.

Control

Chemical treatment cannot efficiently protect the trees against the disease. Contamination can be limited by applying a suitable antiseptic healing paste on injuries caused either by pruning, insects or, more importantly, cultivation tools, as such injuries create wounds where the pathogen can penetrate the sap vessels beneath the bark.

In actual fact, little is known about the epidemiology of this disease. At the time

when it ravaged the plantations of Central and West Africa, very few studies were made of the environmental conditions which were likely to affect its development. On the Ivory Coast, however, it was noted that the first areas to be affected and those that suffered the most had a dry climate which is not the best suited to coffee cultivation.

Recently, in Uganda and in the Democratic Republic of Congo, it was also noted that old and badly maintained plantations were more exposed to the disease. This is, however, very piecemeal information and not much to go on. More serious surveys are needed to clearly define the importance of the effects of the pathogen itself on the one hand and environmental or cultivation conditions on the other.

In the past, attempts were made to prevent the spread of the disease by destroying the affected trees and those in their immediate vicinity either with flame torches or by spraying with carbonylurea to prevent the spores from spreading. It is doubtful that these measures were efficient, as most attempts to eliminate the disease by eradication have failed where cryptogamic diseases are concerned.

Only the careful selection of resistant varieties or clones seems to be the best answer. Important leads, generated by studies carried out in the 1950s on *Canephora*, *Excelsa* and, more recently, on *Arabica* in Ethiopia seem to point towards this solution, and one can but hope that it will provide the answer, particularly at a time when the threat of a serious, widespread outbreak of the disease is again to be feared.

Finally, as in the near future extensive replanting will have to be carried out in Uganda and in the Democratic Republic of Congo, it is to be hoped that this will provide the opportunity for a significant transfer of technology to rural areas so

that replanting and maintenance can progress according to the new rules generated by research.

4.2.3.2 Canker or “Machete Disease”

French	Cancer du caféier
Spanish	Mal de machete, Llaga macana, cancer
Portuguese	Cancro

This affliction, common to coffee, cocoa and various citrus trees, is caused by a fungus, *Ceratocystis fimbriata* (Ellis and Halst.) which decays the woody tissues of the trunk from the collar to a height of about 1 m. The external symptoms are the withering of the leaves, with dead leaves remaining attached to the branches. In the beginning this may only affect part of the canopy, but shoots can also wither.

This disease owes its name to the machete, a tool which often causes injuries to the tree which lead to the development of canker. In actual fact, however, canker is more frequently caused by insects, particularly *Xyleborus*, which bore the holes where the fungi develop in the wood.

Again, apart from careful use of the machete to avoid injuries and the application of a suitable antiseptic healing paste, little can be undertaken to fight the disease. Residual wood should be burnt to prevent the multiplication of the *Xyleborus*.

4.2.3.3 Phloem Necrosis

French	Déperissement à phytomonas du caféier, Nécrose du phloème, Nécrose du liber
Spanish	Necrosis del floéma
Portuguese	Nécrose do floema

As previously mentioned, this disease is not actually a cryptogamic disease.

The plant withers and dies because a flagelliform protozoan, *Phytomonas leptovaorum* (Stahel), develops in the vessels of the phloem causing necrosis of the tissues. This disease only prevails in the northern reaches of South America, Guyana and Surinam.

In this case, however, climatic and environmental conditions are very likely to be more significant factors for the development of the disease than the protozoan itself. In these areas, coffee trees are planted in very humid zones where water frequently floods the soil and drainage is not always sufficient. The submerged roots are asphyxiated by water, and consequently the plant withers and dies. Thus, the presence of the flagelliform protozoan parasite is more likely to be due to the fragility of the tree rather than a pathogen causing the disease. One can only recommend coffee cultivation culture in the most suitable conditions possible.

4.2.3.4 Foliage Rust

Two widespread foliage diseases caused by Uredinals of the Puccinia family, Orange Leaf Rust, also known as Coffee Leaf Rust (CLR), and Grey Leaf Rust are discussed below.

4.2.3.4.1 Coffee Leaf Rust

French	Rouille orangée
Spanish	Roya herrumbre
Portuguese	Ferrugem do cafeeiro
German	Kaffeerost

Causal organism

Hemileia vastatrix (Berk. and Br.).

Historical and geographic survey

Identified for the first time in 1861 in Ceylon (now Sri Lanka), this disease was long considered to prevail exclusively in Asia and

Africa. By a pure fluke, a Brazilian phytopathologist, Arnaldo Gomes Medeiros, identified its presence in a small, more or less abandoned, Arabica coffee plot, in the midst of a cocoa plantation in the State of Bahia in Brazil in 1970. Since then, the disease is known to be prevalent in most coffee-producing countries worldwide. It may seem surprising that the presence of *H. vastatrix* in Brazil was only identified by chance; however, it must be remembered that, in Bahia, at the time, the majority of plantations were given over to cocoa growing, not coffee. Coffee plantations were scattered and did not have any significant impact on the economy of the state. This means that producers and agronomists alike focused on cocoa plantations rather than coffee.

The end result was that once the disease was detected, it became apparent that approximately 50 000 km² of coffee plantations were already contaminated. It would seem, therefore, that the pathogen had been present in Brazil for a long time before it was initially detected [34].

CLR is considered to be the most severe foliar coffee disease known to date. This may be due to the fact that it led to the abandoning of coffee growing in Sri Lanka only 6 years after the disease was first identified. Coffee cultivation was then replaced by tea plantations and this factor largely contributed to the widespread consumption of tea in the UK. At the time, of course, no treatment was available to stem the spread of cryptogamic diseases, not even the famous “Bordeaux Mixture”. The fact remains, however, that this disease is severe and its impact on the economy is most significant, particularly as Arabica plantations are the most vulnerable and they represent four-fifths of the world’s coffee production.

In South America, prior to its detection in Bahia, the merest threat of CLR was liable to generate real panic, as already described. This is, of course, quite understandable as Latin American countries are the major producers of Arabica coffee and, until recently, all their plants stemmed from one specific plant reputedly vulnerable to CLR. On the other hand, the panic reaction was also irrational as it has long been a recognized fact that climatic and environmental factors greatly influence both the impact of the disease on the plant and its potential evolution to epidemic proportions. On the American continent, like elsewhere, climatic diversity means that, although the pathogen is present wherever coffee is grown, the extent of its onslaught ranges from a major disaster to a relatively minor affliction.

Generally speaking, *Coffea arabica* L. is the most receptive of all *Coffea* species to CLR, but, again, receptivity can be more or less active depending on the different varieties.

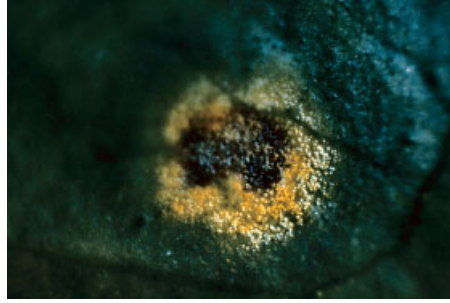
On the other hand, receptivity has also been noted even in coffee plant species reputedly resistant to CLR, like *C. liberica*, *C. excelsa* or *C. canephora* Pierre, as well as in different varieties of these species like Robusta or Kouillou. This can be explained



Picture 4.10 *Hemileia vastatrix*: leaves of *C. arabica* affected by CLR. View of upper and lower sides of the leaf. From: A. Autrique.



Picture 4.11 *Hemileia vastatrix*: close-up of a CLR spot proliferating on a *C. arabica* leaf. From: A. Eskes – CIRAD.



Picture 4.13 Spots of *Hemileia vastatrix* (yellow) surrounded by *Verticillium lecanii* parasiting the spores of *Hemileia vastatrix*. The black blotch in the center is composed of necrotic tissues of the coffee leaf. From: R. A. Muller – CIRAD/IRCC.



Picture 4.12 *Hemileia vastatrix*: leaves of *C. canephora* var. *robusta* affected by CLR. From: J.N. Wintgens.

The spots on the leaves can grow to a diameter of approximately 2 cm and, as their nucleus on the coffee leaf decays, they eventually cover significant areas on the limb. The development of a white pulverulent fungus, *Verticillium lecanii*, which parasites the spores of *Hemileia* is frequently observed.

The feeders generated by the uredospores of *H. vastatrix* penetrate the stomata of the leaves where they form a blister, shaped like an anchor, which develops an abundant, ramified intercellular mycelium.

The mycelium colonizes the tissues of the leaf and, from there, can spread in any direction.

by the allogamy of these varieties issuing coffee types with different characteristics.

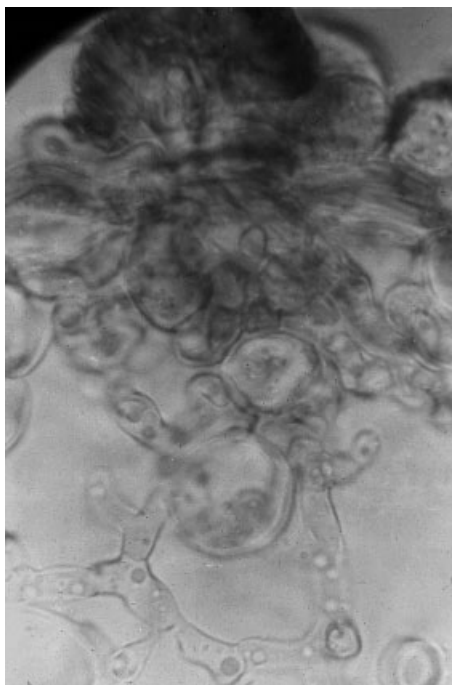
Symptoms

The first symptoms of the disease are small, discolored spots which develop on the underside of the leaves. Within 10 days these small spots increase in size and are powdered with the spores of the pathogen ranging in color from yellowish-orange to bright orange. These cuneiform (wedge-shaped) uredospores have spines along half of their back, hence the name “*Hemileia*”, which means “semi-smooth”. To date, no other host of propagation has been identified for this parasite; only coffee plants seem to be receptive to *H. vastatrix*.

Damage caused

CLR brings about the loss of physiological activity in the affected parts of the limb and causes the leaves to fall. Potent onslaughts of the disease can cause branches to wither completely. This weakens the plant and hinders, or even stops, their development. Frequently badly diseased and weakened coffee trees do not survive.

Studies made on the loss of physiological activity of the leaves by the IRCC and CIRAD in the Centre d’Etudes Nucléaires



Picture 4.14 Longitudinal cut through a coffee leaf showing the abundant and ramified mycelium of *Hemileia vastatrix*.
From: R. A. Muller – CIRAD/IRCC.

de Cadarache have ascertained that at least 20 % of the leaves have to be affected before any significant disruptions can be detected in the photosynthesis or the photorespiratory system of the plant. The effects of the onslaught, however, are not limited to the current crop as they have also been seen to affect the following crop by reducing the production of young wood and diminishing stock-piling in the plants [6].

On the other hand, the real damage caused by the onslaught of the disease is assessed in terms of a loss of yield; nevertheless, the level of infected foliage should also be used as an indication of the damage inflicted. To date, however, no real correlation has been established between the extent of the foliar damage and its impact

on the yield. No precise data exists as yet which would enable a clear prediction of the extent of damage to the potential yield. It is generally assumed that 1 % of damaged foliage can lead to a yield loss of 1 % in the following year.

In Brazil, it is thought that, depending on the location and the year, the disease can cause losses which vary from 25 to 45 %.

On the basis of general observation, one could come to the conclusion that, essentially, it is chiefly when young leaves are affected that the disease inflicts the most serious damage, as older leaves can be considered to have already lost most of their potential for physiological activity and are due to fall anyway.

Propagation

The spores only germinate in damp conditions, in the presence of water. Epidemics are, therefore, prevalent during the wet season. Rainy spells show an increase in the spread of the disease and periods of intense infection correspond to those of high rainfall.

Germination of *H. vastatrix* spores is also more active at temperatures between 20 and 25 °C, showing its highest activity at 22 °C [38]. At temperatures below or above this level, germination is either slow or nil. The temperature is, therefore, a further important factor to take into consideration.

Another element is the fact that the germination of spores has been observed to take place chiefly in the dark, at night [42].

These observations do not necessarily mean that germination never takes place during the day or in cooler temperatures as, hypothetically, the disease could propagate in the centre of the foliage, particularly under shade trees or on a cloudy day when daylight scarcely penetrates the canopy.

As temperatures are directly related to altitude, the risk of infection can be defined on the basis of this parameter. If, in equatorial zones, the onslaught of the disease is more intense at low altitudes, it will be less so higher up and, in fact, practically nil above the 1300 m level. Nearer the tropics, the danger level has been drawn at the 800–900 m level [2].

The humidity generated by the presence of shade trees is generally thought to be favorable to the propagation of the disease with a ratio of intensity that increases with the density of the canopy. This “hands-on” observation has, however, been countered by research carried out by CIRAD/IRCC in France which has revealed that sunlight can increase the plant’s receptivity to the disease.

Furthermore, it has also been ascertained [20] that coffee trees, induced to high yield crops by exposure to the sun, are more receptive to *H. vastatrix* [6, 30, 50].

Significantly, it has been noted, too, in both Cameroon and Guatemala, that plants of the same genotype can be receptive to the disease when in the fructification stage and practically immune when in the purely vegetative stage like, for instance, just after stumping. This cannot only be explained by the lack of inoculum which is generated by the leaves of the preceding year.

Physiological observations of this type are worth following up with careful research as they may lead to interesting developments where corrective nutritional measures, like increasing macro- or oligo-elements during the yielding period, can be used as a means to increase resistance to the disease.

Surveys carried out in Honduras by Avalino between 1994 and 1997 have actually established a link between shading, yield and the intensity of CLR infection. Sche-

matically, one can now say that a coffee plantation with a relatively low yield, with or without shade, will only suffer a slight onslaught of the disease, whereas for a plantation where the yield is high, the degree of infection will be more intense in proportion to the density of the shade. In other words, the denser the shade, the more intense the infection will be.

The same author has also noted that a low soil pH combined with a high yield can also predispose the coffee trees to the disease. A similar relationship between soil pH and CLR intensity has also been found in New Caledonia [39], although these observations have yet to be substantiated.

H. vastatrix spores, like the spores of other rust diseases, can be carried by the wind over long distances. It has been assumed that spores have been carried by wind across the Atlantic from Africa to Latin America by high altitude currents [12].

Although *H. vastatrix* spores can be carried by wind or by any other factor that moves within the plantation, including people who are active vectors, basically, in any given plantation, the chief source of contamination at the beginning of the infection is to be found in the infected leaves of the preceding campaign. These leaves host the pathogen so that, when the first rains generate the new vegetation, the fungi are also activated to produce new uredospores [35]. Contamination then trails from leaf to leaf and accelerates as the infected spots multiply the production of spores. As residual intrinsic inoculum indisputably plays a major role in the start-up of the infection, the use of a selective defoliant should be considered. This selective defoliant should be geared to target old leaves or leaves that have been previously infected; consequently, their elimination

would also destroy the pathogen which would delay and reduce the onslaught of the disease. Experiments along this line that were initiated in Cameroon 30 years ago should be re-activated and developed. Such measures, which aim to destroy the residual inoculum left over from the preceding year, would be all the more successful if they were extensively implemented. If all the plantations in a given zone were treated the same way, the early effects of a residual inoculum strong enough to launch the disease would be avoided, contained or, at least, delayed.

In some areas, like, for instance, the Bamoun in Cameroon, natural phenomena have this very effect. Coffee plantations in this area, situated at an altitude of approximately 1100 m, suffer from drought, aggravated by an ashy-porous soil. This causes the trees to lose practically all their leaves by the time the first rains set in with the result that, due to a lack of residual inoculum, the onslaught of the disease is significantly delayed and little or no damage affects either the ongoing crop or future crops.

An epidemiological survey carried out by in areas with a very dry season in the south of the Chiapas in Mexico drew similar conclusions [2].

Another alternative to the use of a selective leaf-killer would be the use of systemic fungicides to destroy or reduce the residual inoculum on old leaves [3, 27]. This method would have the added advantage of preventing the propagation of the pathogen while maintaining the old leaves so that they can still play their original role by transferring nutritional elements to the developing leaf tissues.

Chemical control

As the significant damage caused by disease has been universally recognized for the last 30 years, research has been actively

attempting to develop methods for controlling it.

Chemical control with copper-based products has proved to be efficient if applied at regular intervals at least twice and up to 5 times a year, depending on the area. The first application should take place just after the first rains and subsequent applications should be made approximately every month.

The most widely recognized of the modern fungicides is Triadimefon. Its systemic action, combined with preventive, healing and anti-sporulant properties, generates good results with fewer applications. Other fungicides of the triazole group like, for instance, Cyproconazole or Hexaconazole, have also been successful. Fungicides of this type can complement copper based fungicides and be used alternately.

No rigid rules can be established for the application of fungicide treatments. Epidemiological surveys substantiated by precise experimentation should be established for each zone. They should indicate the sequence, length and number of applications recommended for the area. Treatment should start with the first rains, as soon as possible before the first attacks appear, and should continue for as long as the pathogens are active.

Obviously, in areas where a very dry season has caused the fall of many infected old leaves, the lack of residual inoculum will cause the pathogen to develop far later. In this case, if chemical control is still necessary, it can be initiated at a later date and spaced out accordingly. It may not even be necessary at all as chemical treatment carried out when the peak of the infection comes about at the end of the berry-ripening period is of very little use and, from an economic point of view, scarcely to be recommended.

On the other hand, treatment should not be stopped too soon. A delayed onslaught of the disease may still infect the leaves and cause a physiological weakening which would affect the transfer of nutritional elements to the new shoots due to carry the following year's crop. This further highlights the need for specific surveys to monitor chemical control in each area.

Genetic control

Specific or "vertical" resistance

Solutions for genetic control have been, and are still being, actively researched.

Operational results have already been ascertained, but care needs to be taken to maintain them and research still has to develop replacement solutions to be implemented in cases where the pathogen has managed to break down the defenses introduced genetically.

Following initial research [32], which now dates back considerably, further research carried out at Oeiras in Portugal

by the Centro de Investigaçao das Ferugens do Cafeeiro (CIFC) along the lines of specific or vertical resistance has identified virulence factors in the pathogen [45]. To date, 40 distinct physiological races of *H. vastatrix* have been recognized as carriers of one or several virulence factors.

The specific resistance mechanism is illustrated in Tab. 4.3 where the virulence factors of the parasite (v) are portrayed by keys and the resistance factors in the host plant by padlocks (SH). Where there is compatibility between a key and a padlock, vulnerability has been established, and where there is no compatibility between a key and a padlock, the opposite resistant phenomenon is noted.

Resistance factors SH6 to SH9 (or SH10) issued from *C. canephora* and found in Timor Hybrids (spontaneous hybrids generated between *C. arabica* and *C. canephora*) are especially interesting.

Timor Hybrid 1 (832/1) represents resistance group "A", and resists to all known

Table 4.3 *H. Vastatrix*: chart illustrating compatibility (sensitivity) and incompatibility (resistance) between CLR and the coffee tree

		v5		SH5		
Race II			→			Group E
	V2	v5		SH2	SH5	
Race I			→			Group D
	V2	v5		SH5		
Race I			→			Group E
	V1	v5		SH5		
Race III			→			Group E
	V1	v5		SH2	SH5	
Race III			↗			Group D
		v5		SH2	SH5	
Race II			↗			Group D

→ = compatibility (sensitivity).

↗ = incompatibility (resistance).

From: J. Avelino – CIRAD-CP.

varieties of the pathogen and carries at least four resistance factors not to be found in SH1 to SH5.

Timor Hybrid 2 (1343/269) represents resistance group “R” and carries at least two resistance factors not to be found in SH1 to SH5. It is resistant to all known types of the pathogen except eight.

Other types of the Timor Hybrid with different genetic components represent resistance groups 1, 2 and 3, and resist to all known races of the pathogen with a few recognized exceptions.

To date, this specific research carried out by the Portuguese team of Oeiras and a Brazilian team based at the University of Viçosa has led to the development of the “Catimor”, a coffee hybrid issued from a cross of Caturra (a dwarf Arabica Bourbon very vulnerable to leaf rust) and the Timor Hybrid. Colombian research has developed the “Colombia” and breeders in Kenya have developed the “Ruiru 11”. These varieties are now available for use in coffee plantations and offer significant resistance to leaf-rust diseases. The same applies to the “Icatu” developed in Brazil, which is an Arabusta (an artificial hybrid from Arabica and Robusta) back-crossed to Arabica without using the Timor Hybrid. It carries the resistance genes found in Robusta and offers a good genetic resistance to CLR.

The introduction of resistant varieties issued from the Timor Hybrid should be implemented with all due precaution, bearing in mind the fact that *H. vastatrix* can be easily mutated and that diversity in coffee genotypes stimulates genetic diversity in the pathogen. This is due to the fact that mutants of the pathogen may well find genotypes which suit them in the diversified varieties of coffee trees and will propagate all the more easily. Several indices have been noted which substantiate this theory.

As to the performance of Catimor plants in the field, producers are not unanimous. Many strains have been developed in an attempt to parry the rapid exhaustion of the plant (called “agotamiento” in Central America) which can appear after the first main crop. Small-scale trials are, therefore, recommended prior to committing the entire plantation to the Catimor variety.

To date, the best strains of Catimor are the T5175 and the T8667, but even the T5175 can show serious physiological failures under certain climatic conditions, particularly at low altitudes, in sites where a lengthy drought is followed by heavy rains. The major recognized affliction is known as “black bean” and can affect 80% of the production. In this case, although the berries appear to be healthy, the beans are black and withered.

Moreover, some of these resistant commercial varieties are reputed to have off-tastes, mainly when the yields are very high. In the current crisis period which has led prices to drop to an all time low, quality is a means of obtaining higher prices. Thus, resistance and high yields are not enough. Today, the real challenge of research is to create resistant varieties whilst preserving cup quality.

General or “horizontal” resistance

To hold a full hand of aces, however, one cannot bank solely on specific resistance as described above. While, at present, the Catimor, Colombia and Icatu offer the most advanced and efficient material, it may not be the case on a long-term basis, particularly as, due to mismanagement, these breeds are already losing genetic resistance.

It is, therefore, advisable to examine the possibilities of a different method of genetic control, by associating “general” or “horizontal” resistance.

Where “specific” resistance reacts qualitatively, in other words the trees are either victims of an onslaught of the disease or they resist it, “general” resistance reacts quantitatively and will only be noticeable once the disease has actually taken hold. Consequently, “general” resistance aims at reducing the intensity of the onslaught and lengthening the period of latency, thus reducing sporulation. Observations made on the reactions of several genotypes to the infection substantiate this theory.



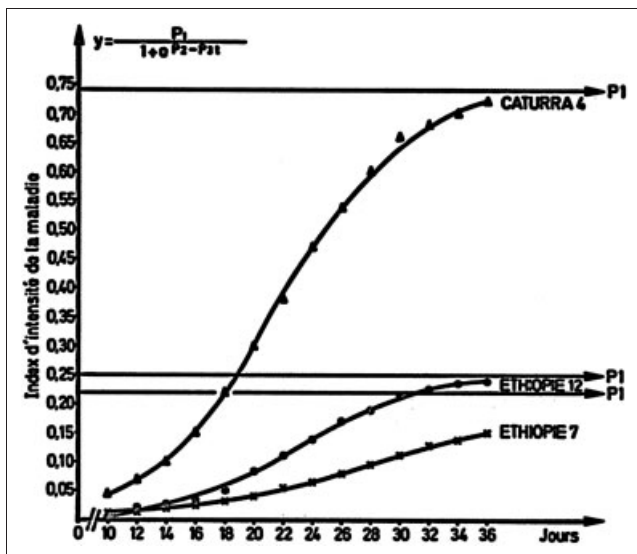
Picture 4.15 Evidence of “general” resistance to *H. vastatrix* between Catimor and susceptible Arabicas in the field. From: J. N. Wintgens.



Picture 4.16 Evidence of “general” resistance to *H. vastatrix* in *C. arabica*. For inoculations made with the same pathogen strain under the same conditions, the intensity of the disease is variable depending on the genotype of the coffee. From: D. Bieysse – CIRAD-CP.

Picture 4.15 clearly illustrates significant differences in the reactions of different genotypes to standard inoculations made on the same date.

As all the aspects of the resistance can be measured, the results can be determined globally under the heading “Intensity Index”. This enables straightforward comparisons between the various genotypes observed.



Picture 4.17 Expression of the “general” resistance to *Hemileia vastatrix*. Comparison of the evolution of different intensities (Intensity Index) of *H. vastatrix*, race 2 on three distinct genotypes of *C. arabica*.

Recent research carried out at CIRAD-CP has revealed that this type of resistance exists naturally in Arabica, particularly in certain wild genotypes to be found in Ethiopia, in “vulnerable” varieties of Cane-phora and in Catimor genotypes which have lost their “specific” Timor Hybrid resistance factors [23, 26, 28]. Earlier, but less in-depth, research carried out in Brazil had already discovered this type of resistance both in *C. arabica* and in varieties issued from the Timor Hybrid [13, 14, 17–19, 52].

To date, research carried out on this type of resistance has been promising, but has not yet been developed sufficiently to justify implementation. A few questions remain unanswered, particularly the fact that, although distinct evidence of “general” resistance can be noted in young coffee trees, it seems to dwindle as the tree grows older. Added to which, fructification increases the “vulnerability” of the coffee tree to the pathogen.

Research should be carried out to examine the real potential of “general” resistance and the methods to be set up for its implementation as it may well be the best answer in cases where “specific” resistance is either lost or greatly reduced. The two methods of genetic control could also be complementary as “general” genetic control could be used to back-up “specific” genetic control and ensure its efficiency for a longer period.

It should also be remembered that the quality of the coffee produced from genotypes issued from the Timor Hybrid is not fully satisfactory; this criteria would justify the need to introduce pure Arabica leaf-rust resistant varieties issued from the Ethiopian genotypes.

4.2.3.4.2 Grey Leaf Rust

French	Rouille farineuse, Rouille grise
Spanish	Roya harinosa

Causal organism

Hemileia coffeicola Maublanc and Roger.

Historical and geographic survey

Noticed for the first time on Arabica in the Cameroon by Coste in 1929, Grey Rust was given its name by Maublanc and Roger. In 1953, Muller first noted its appearance on Robusta, again in Cameroon. Later on, Sacas noted that it was also present on Excelsa and *Coffea de la Nana* in the Republic of Central Africa. Grey Rust spreads slowly and, to date, is only prevalent in Central Africa (Cameroon, Central African Republic, Nigeria, Sao Tome and Angola). In Togo, it has been remarked on a Robusta clone by Partiot and, in Ivory Coast, by Lourd on certain varieties of wild coffee. In the latter case, however, although this has yet to be checked, there is some doubt as to whether it might not be another, morphologically similar, type of the disease.

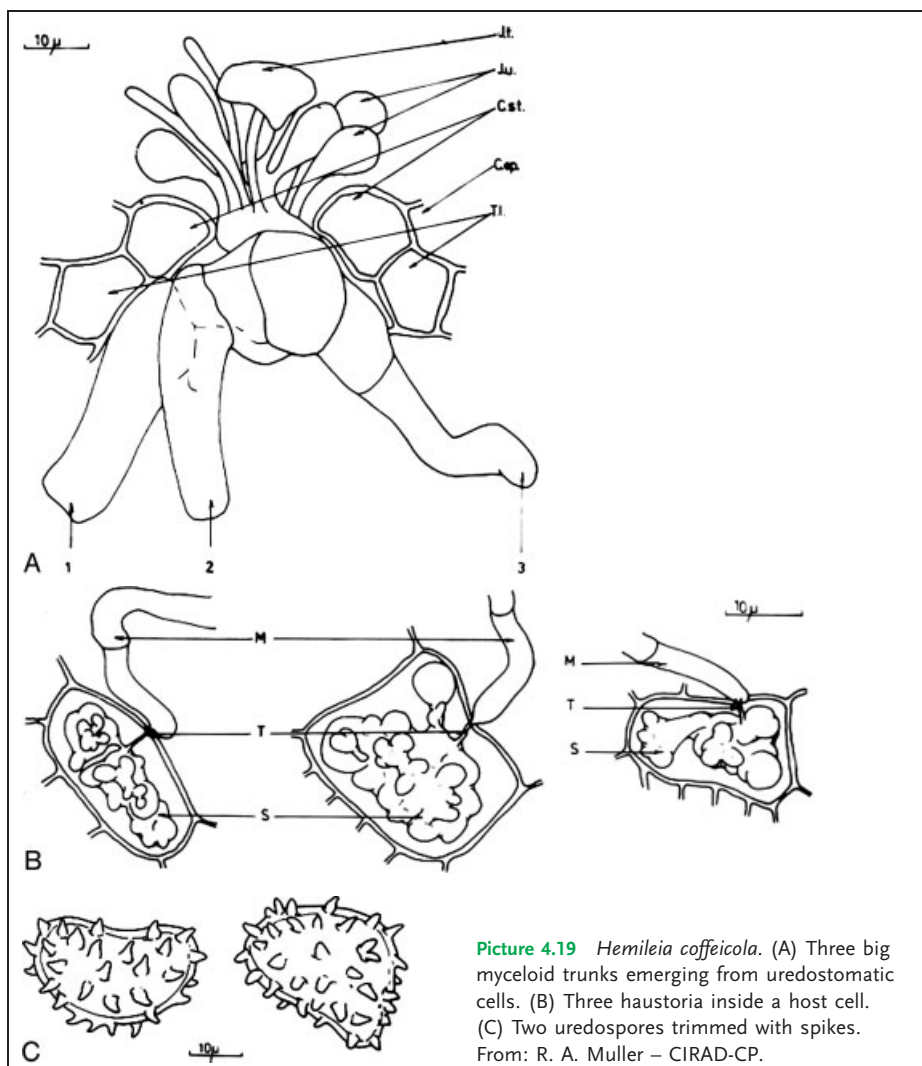


Picture 4.18 Comparison of the symptoms of attacks of *Hemileia vastatrix* (right) showing clearly defined spots and *Hemileia coffeicola* (left) showing unevenly spread blotches on the leaf surface. From: R. A. Muller – CIRAD-CP.

Symptoms

The macroscopic and microscopic characteristics of Grey Leaf Rust make it easy to distinguish from CLR. This is chiefly due to the shape of the spots generated by *H. vastatrix* and *H. coffeicola*, respectively. Where *H. vastatrix* spots are clearly defined and contained, *H. coffeicola* shows uneven blotches spread on the underside of the leaf. The leaves do not lose their color, they are just covered by the yellow spread which is the first visible symptom of the disease.

From a microscopic point of view, the mycelium inside *H. coffeicola* is not abundant with a number of tiny haustoria like *H. vastatrix*. On the contrary, it generates only two or three large intercellular filaments with no ramifications. Each filament is terminated by a large globulous and complex sucker which invades the cells of the host. The spines on the back of the uredospores of *H. coffeicola* are also better defined than those of *H. vastatrix*.



Picture 4.19 *Hemileia coffeicola*. (A) Three big myceloid trunks emerging from uredostomatic cells. (B) Three haustoria inside a host cell. (C) Two uredospores trimmed with spikes. From: R. A. Muller – CIRAD-CP.

Damage

From a macroscopic point of view, Grey Leaf Rust is less visible and, therefore, less easy to detect than CLR. Consequently, planters tend to minimize its importance. In Cameroon, however, surveys have shown that Grey Leaf Rust disrupts photosynthesis and the assimilation of phosphorus and that it causes considerable loss of leaves, although not quite as much as the loss of leaves caused by CLR. It can also be a factor which seriously curtails production as intense onslaughts of the disease can either significantly reduce or totally destroy the crop yield of the following year [35].

Propagation

As, geographically speaking, *H. coffeicola* is relatively localized and its discrete symptoms are not taken too seriously, this disease has not been extensively studied except in Cameroon. Like *H. vastatrix*, *H. coffeicola* propagates fastest in wet weather and its annual development depends on

rainfall. Surveys carried out in Cameroon at high altitudes reveal that the first symptoms appear approximately 1 month after the first rains and that the development of the epidemic is rapid. The peak of the attack on coffee leaves can be very high towards the middle or the end of the rainy season.

Apart from the fact that the rainy season favors the development of *H. coffeicola*, no other specific environmental conditions seem to be required for the disease to spread. While *H. vastatrix* does not proliferate at high altitudes, *H. coffeicola* can be as dangerous to high-altitude plantations as those at lower altitudes. It has been abundantly detected in high-altitude plantations as well as on Arabica and on “susceptible” Robusta genotypes right down to sea level (Tab. 4.4). This clearly indicates that Grey Leaf Rust is a potential danger to coffee cultivation worldwide and coffee producers should be more aware of the threat of this pathogen.

Table 4.4 Chart showing the degree of virulence of *Hemileia coffeicola* according to altitude

Species	Locations	Cultivars	Years	Aggression rate (%)
C. arabica	High altitude, 1650 m Babadjou	Jamaica	1970	14.82
			1971	22.42
			1972	37.16
			1973	16.81
	Low altitude, 1110 m Foumbot	Jamaica Bourbon Mayaguez Mulungu 5 Ethiopia 37 (217/9)		12.84
			1974	36.52
				39.65
				38.11
C. canephora var. robusta	Hot zone, 175 m (Barombi-Kang)	B 49 C 1 B 21 N 3	1974	38.87
				36.52
				31.68
				28.15

From: Tariot and Lotode [52].

Chemical control

Chemical control is possible. Copper-based products can be efficient and the most efficient is cuprous oxide (Red Copper) with a 50% Cu content and used as a brew at 0.5%. In Cameroon it has been shown that three treatments per year at monthly intervals are enough to control the onslaughts, even on irrigated plantations. At high altitudes, it has also been noted that cupric treatments used against CBD are also efficient against Grey Leaf Rust [35].

Triazoles, used to control *H. vastatrix*, can also be used to control *H. coffeicola*.

Genetic control

Little is known about genetic control of Grey Leaf Rust but, according to observations made in Cameroon by [52], it would seem to be possible. A collection of Arabica including approximately 100 of Ethiopian origin has have been classified and compared according to the level of their resistance in the field. It is also evident that the *Canephora* species hold genotypes which offer different levels of “susceptibility”. On the other hand, apparently, genes that resist to *H. vastatrix* would not necessarily be the same as those that resist to *H. coffeicola*.

4.2.3.5 American Leaf Spot

French	Petite vérole, Maladie américaine des feuilles
Spanish	Ojo de Gallo, Gotera, viruela, Mancha americana
German	Amerikanische blattkrankheit
Dutch	Stilbella-ziekte

Causal organism

Omphalia flavida (Maublanc and Rangel) = *Mycena citricolor* (Berk. and Curt.) Sacc.

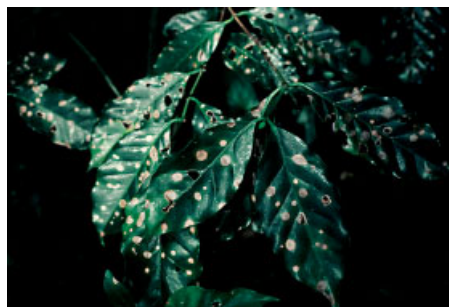
Historical and geographic survey

This disease is essentially prevalent in Latin America; more specifically, in Costa Rica, where it has a significant impact, and in the Caribbean. It affects all the organs, stems, branches, fruit and leaves of all cultivated coffee trees. Its onslaught is not limited to coffee trees as it also attacks a number of plants of the family of the Leguminosae and Compositae as well as a few adventitious plants, which could well be the fungi banks, representing a potentially active source of contamination for coffee trees.

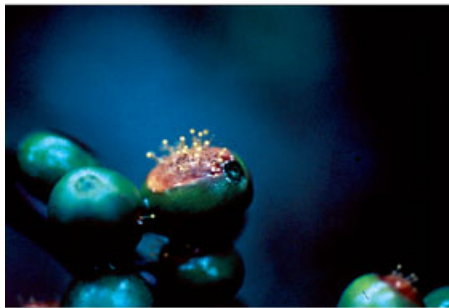
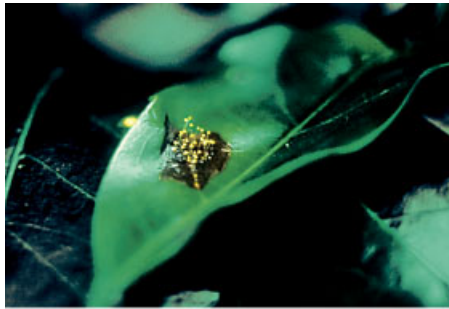
Symptoms

American Leaf Spot is recognizable by its numerous of round spots which appear on the limbs of the leaf. The tissues of the spots wither and fall out, leaving a hole in the leaf which, eventually also falls. As a general rule, these round spots vary in color from chestnut brown to a deeper brown, but other types of spots, reddish in color and more jagged in shape, have also been detected.

The Basidiomycete from the Agaricaceae order, named *Mycena citricolor* in 1950 by Dennis, is the responsible pathogen. It has two types of fructifications: vegetative and sexual. The vegetative fructification looks like a series of little headpins a few



Picture 4.20 *Mycena citricolor*. View of an attack on a coffee leaf. From: J.N. Wintgens.



Picture 4.21 *Mycena citricolor*. Asexual fructifications on a coffee leaf and fruit shaped like small headpins. From: Sandoz, Guatemala.

millimeters high, and the sexual fructifications are carpophores with a diameter of a few millimeters and 1 cm high. The latter are yellow and of a mossy consistence, but as they usually appear on fallen leaves, they are not believed to be active in spreading the disease.

Propagation

A survey carried out in Guatemala by Avellino [6] reveal that, in this country, near to the tropics, American Leaf Spot is particularly serious at high altitudes, above 1200 m. Its development relies on very damp conditions and chiefly prevails in plantations where bad maintenance is combined with excessive shade. On the other hand, it also prevails in the absence of shade, but in specific environmental conditions, at high altitudes where the weather is often cloudy or foggy and thin, persistent rains are frequent.

The disease can cause losses of up to 35 % of the expected yield over 2 years [7]. These losses are due to the infected fruits which drop and to the negative impact of the disease on the growth of the branches which, in turn, affects the following crop.

Chemical and cultivation control

For many years this disease was controlled by spraying lead arsenate, but this practice is no longer authorized so growers have resorted to more traditional, copper-based fungicides like the “Bordeaux Mixture” which is also proving to be efficient [5]. It is worthwhile noting that the calcium phase of the “Bordeaux Mixture” actively controls the disease by preventing the development of oxalic acid, the toxin generated by the fungi when penetrating the leaf of the plant [40]. Modern triazoles with systemic effect like Cyproconazole or Hexaconazole can also be used, either on their own or alternating with the cupric fungicides, in order to reduce any residual inoculum after an onslaught which would otherwise generate sources of contamination for a further onslaught.

Correct maintenance methods like weeding, pruning and shade control are also necessary as they either prevent the onslaught of the disease or, at least, reduce

its intensity. A well-ventilated plantation and stumping in alternate rows will offer fewer possibilities for the disease to develop and spread. Surveys carried out in Guatemala have shown that the percentage of diseased leaves on adult coffee trees in a row next to a stumped row could be as little as 15 %, whereas the same trees in a row next to another non-stumped row of adult trees could have up to 45 % of diseased leaves [5].

Genetic control

As American Leaf Spot originated in Latin America and coffee plants were initially imported from Africa, their native continent, there can be no parallel evolution between the host and the pathogen. There is, therefore, little chance of developing “specific” resistance. This means that possibilities for “general” resistance would need to be examined. To date, very little research has been carried out along these lines, although comparisons between Catimor and Catuai, both inoculated cleanly, have pointed to different levels of resistance to both the penetration and the fructification of the disease. Catimor appears to be more resistant to penetration and Catuai more resistant to its fructification [37].

Resistance of this type, however, tends to disappear as the fungus adapts to the new environment. For more conclusive results, the scope of investigation would need to be extended to more varied genotypes, including Ethiopian Arabica or interspecific hybrids. On the other hand, different levels of field resistance could be pinpointed by taking the structure of the plant into consideration – the more compact the plant, the higher the contagion and the greater the development of the disease within the plant itself.

4.2.3.6 Blister Spot

French	Maladie des taches huileuses
Spanish	Mancha mantecosa, Mancha aceitosa
Portuguese	Mancha manteigosa

The economic impact of this disease is relatively low, but it is mentioned here because specialized literature highlighted it in years gone by F. L. Wellman in 1961 and 1972. It remains, however, localized in America and is characterized by small light green, greasy-looking translucent spots which appear on the leaves. These spots, which are slightly bulbous in appearance, spread over the leaf and link; when this occurs, they decay and causes the leaves to fall which, in turn, causes the branches to wither. Young leaves are particularly vulnerable. Frequent attacks lead to the decay and the death of the entire tree. This disease is to be found mostly in nurseries. Initially, it was thought to be generated and propagated by aphids (*Toxoptera auratii*), but more recent studies seem to point to a cryptogamic disease generated by *Collectotrichum* sp.

In 1991, Matiello mentioned it as being especially virulent in Conillon of the *C. canephora* variety where, in Brazil, up to 15 % of the plants were exposed. Icatu would seem to be less vulnerable and Arabica very rarely attacked by this disease in Brazil. In the state of Vera Cruz, in Mexico, however, Blister Spot infected a high percentage of Arabica plants.

Chemical control is possible, but considered, in Brazil, to be too costly as several repeated treatments with systemic fungicides, like Benlate (no longer authorized), would be needed to ensure efficient control. Other methods would, therefore, need to be researched and implemented as the economic impact of the disease would justify the cost of the research.

4.2.3.7 Brown Eye Spot or *Cercospora* Blotch

French	La maladie des yeux bruns, Cercosporiose
Spanish	Mancha parda, Mancha de hierro
Portuguese	Mancha do olho pardo
German	Braunaugenkrankheit

Causal organism

Cercospora coffeicola (Berk. and Cooke).

General comments

This disease attacks the leaves of coffee trees of all ages, but its impact is the most serious on young coffee plants both in germination beds or in nurseries. The infected plants may lose the majority of their leaves, or even all of them, and the berries can also be diseased. Arabica trees seem to be the most vulnerable.

The symptoms which appear on the leaves are round, grey-brown spots approximately 1 cm in diameter and with a lighter-colored dot in the center.

Concave pressure points, known as “thumb dimples”, appear on the berries. These marks are brown and covered with a visible grayish fuzz which is the fungi fructification. Frequently, these marks are rimmed in red and the surrounding fruit pulp tends to ripen prematurely.



Picture 4.22 *Cercospora coffeicola* – Brown Eye Spot. Attack on coffee leaves. From: A. Autrique.

As a general rule, the coffee trees which show these symptoms are more yellow than green because this disease is, typically, a disease due to a physiological deficiency within the plant – the fungi only develop as a result of the general condition of the plant. Studies carried out on Arabica in Cameroon and in Colombia and on Arabusta in the Ivory Coast have revealed that the principal causes of this disease are a lack of nitrogen and, possibly, potassium as well as an over-exposure to sunlight. The lack of essential minerals is aggravated by sunlight as the sun activates production regardless of nutritional deficiencies. It also activates the growth of grasses which are known to be greedy consumers of nitrogen [35].

Consequently, obvious methods for fighting this disease reside in correct maintenance of the plantation as a whole, with special care given over to weeding out the seedling beds and the nurseries. Nitrogen fertilizers are also recommended. Fungicide treatments may reduce the presence of *Cercospora* but they do not solve the basic problem.

4.2.3.8 Other Smudgy Foliar Diseases

4.2.3.8.1 Coffee Canker

French	La lèpre du caféier
Spanish	Lepra del cafeto

This is caused by green algae, *Cephaleuros virescens*, which develop on the leaves. Initially reddish-brown and of a velvety texture, these smudges become grayish or greenish and are purely superficial. They are generally due to an excess of humidity and do not cause much damage. In the worst case, they can hamper the assimilation of chlorophyll and are only mentioned here because they are the object of widespread

discussions. They can be compared to lichen which develop on old leaves in wet climates and are no more important.

4.2.3.8.2 Sooty Blotch Mold or Sooty Fungus

French	Fumagine
Spanish	Mancha de moho, Carbon
Portuguese	Fumagina
German	Russtaupilze
Dutch	Roetdauwschimmels

Sooty Blotch Mold is a fungus which develops on the sweet excretions of aphids and cochineal insects, most often *Capnodium coffeae* Pat. As it covers the leaves, the branches and even the berries with a thick, black crusty coating, this disease looks more virulent than it really is. Basically, it chiefly only hampers photosynthesis but, more significantly, it indicates the presence of an unwarranted number of aphids and colichineal insects. Treatment should, therefore, aim at eradicating these unnecessary hosts as well as the ants which transport them.



Picture 4.23 Sooty Black Mold. Observe the whitish Mealybugs and black papery films of fungal growth on leaves and berries. From: Nestlé.

4.2.3.9 Thread Blight Diseases

Traditionally, the different types of Thread Blights were considered to be of minor importance as they were chiefly found on neglected coffee trees or those under exces-

sive shade in a damp environment. More recently, however, Thread Blights have been detected in non-shaded plantations, although at low altitudes and in damp climates in various Central American countries. It has also been noted that they prevail in high density plantations which use dwarf varieties of Arabica, especially Caturra, by maintaining confinement. They have become a real problem in western Guatemala and are also a serious concern in the Chiapas state of Mexico as well as in the Lake Yojoa region and in the Santa Barbara department of Honduras.

The increasing impact of Thread Blights may be due either to the evolution of agricultural practices or to the fact that coffee-growing areas have now extended into zones which favor the development of the pathogen.

The three major Thread Blights are Koleroga or Black Rot, Marasmoid Thread Blight and Horse Hair Blight. All three are caused by Basidiomycetes.

4.2.3.9.1 Koleroga

French	Maladie des fils blancs
Spanish	Mal de koleroga, Moho de hilachas, Infierno del cafeto, Mal de hilachas
Portuguese	Koleroga
German	Silberdrahtkrankheit

Caused by *Pellicularia koleroga* (Cke.) = *Corticium koleroga* (Cke.) Höhn, this disease prevails on every continent and on many varieties of woody plants like citrus, ficus, cocoa, etc. Fine whitish threads develop on the branches and the twigs, and reach the leaves which they cover in a loose web. The leaves wither and die. They then fall off and hang in the web of the fungus.



Picture 4.24 *Corticium koleroga*. (1) Infected leaves and twigs. (2) Dead leaf still hanging on the twig but retained in the web developed by the hypha of the fungus. (3) Cut through the mycelium and the parasite. From: Bayer.



Picture 4.25 *Corticium koleroga*. Infected branch. Observe the whitish web developed on the twig and the leaf. The progress of the disease is centrifugal. From: J. N. Wintgens.

4.2.3.9.2 Marasmoid Thread Blight

French Maladie de la toile d'araignée

Spanish Enfermedad de la telaraña, arañera

Caused by *Marasmius scandens* Mas, this disease is similar in appearance to Koleroga. A fine film of grayish mycelium envelopes the leaves and the twigs, causing the leaves to wither and die. Again they remain hanging within the enveloping mycelium.

4.2.3.9.3 Horse Hair Blight

French Maladie du crin de cheval

Spanish Molo de hilacha

Caused, in this case, by *Marasmius equicrinis* (Muell.), long thin black threads develop on the leaves and the twigs. These threads are capped by tiny fungal carpophores. As in the two previous cases, the diseased leaves wither and die only to remain hanging by the filaments.

The best way to avoid attacks of this type of fungus is, obviously, correct maintenance of the plantation. This entails strict control of shade trees or their removal and careful pruning of the coffee trees themselves with a view to eliminating infected branches. All diseased branches should be cut off and burnt. Chemical control with cupric compounds or other fungicides can be used in extreme cases, but further investigation is needed to pinpoint the most efficient fungicides.

4.2.3.10 Die-back

French Anthracnose des rameaux

Spanish Antracnosis de las ramas, Muerta regresiva

When this disease attacks a coffee tree, the branches wither from their tips down towards the older limbs and the core of the



Picture 4.26 Coffee cherries affected by *Colletotrichum gloeosporioides* (Anthracnose) showing brown concave lesions on ripening cherries. From: G. Castillo.



Picture 4.27 *Colletotrichum gloeosporioides*. The fungus causes the necrosis of the still green terminal tissues of the young twigs. From: A. Autrique.

plant, hence the name “die-back”. Leaves and berries on the affected branches also wither and fall.

Fungi of different varieties have been associated with this disease, but the most common cause is *Colletotrichum coffeanum* Noack which develops on mummifying tissues. This *Colletotrichum* is often associated with *Cercospora coffeicola* as pathological characteristics known in Central America by the all-inclusive name of “chasparria negra”.

It should be clearly noted, however, that this *Colletotrichum* is not to be confused with *Colletotrichum kahawae* Waller and Bridge (ex. *Colletotrichum coffeanum* Noack *sensu* Hindorf) which causes CBD.

In fact, die-back is a physiological disease generated by unsuitable environmental conditions, over-production and inadequate cultural methods.

As a general rule, any circumstances that weaken the coffee tree can lead to die-back. More specifically, these can be one or a combination of the following:

- Over-exposure to sunlight with deficient nitrogen nutrition

- “Hot and cold” shocks due to sharp differences in temperature, during the day and the night
- Difficult acclimatization of certain genotypes
- Over-production with deficient compensation in fertilization
- Exhaustion after the first major yield in some Catimor varieties

- Intense attacks of CLR with loss of leaves and general weakening of the plant
- Inadequate soil maintenance
- Erosion which prevents the soil from retaining moisture and essential nutrients.

Consequently, the most efficient way of fighting die-back entails the practice of sustainable cultivation. In plantations where Arabica is farmed intensively, the entire range of technical support and input available should be correctly implemented. Dwarf varieties, like Caturra, which can be grown densely in full sunlight, need exacting treatment. Fertilization should be adapted to the native soil fertility and rates should adequately support the high yield expected of the coffee tree, particularly where the soil is not fertile enough.

It should also be noted that shade generally protects coffee trees from die-back but, at the same time, it is not conducive to high yields. The more modest yields on shaded plantations do not exhaust the coffee plant to the same extent as the high yields harvested on sun-exposed plantations.

On *C. canephora* var. *robusta*, the die-back *Collectotrichum* attacks the leaves where it generates brown spots rimmed in yellow and induces defoliation.



Picture 4.28 *Collectotrichum coffeanum*.
Attack on the leaves of *C. canephora* var. *robusta*.
From: R. A. Muller – CIRAD/IRCC.

When this serious affection occurs in propagation plots, it stunts the vegetative growth of shoots assigned to the production of cuttings.

Only a limited number of leaf nodes remain on the afflicted plants which decreases their potential for cutting production. This is yet another aspect of physiological weakness, although it has not been possible to determine the exact cause. Some Robusta clones are susceptible when they are very young and seem to develop with no after-effects when they are cultivated at a later date provided they are treated with due care. These symptoms, however, should be taken as pointers to possible bad acclimatization. Susceptible clones should be discarded before they are distributed to farmers as they may never develop into productive trees.

Collectotrichum coffeanum can also cause the small, discolored spots found on Arabica known as “weak spots”. This minor affliction has no economic impact.

4.2.3.11 Pink Disease

French	Maladie rose
Spanish	Enfermedad rosa
Portuguese	Mal rosado

Caused by *Corticium salmonicolor* Berk. and Br., this disease covers the bark of stems and branches with a powder-like salmon-pink coating which pales later on and thickens into a crust. The diseased parts wither and die.

Little is known about this disease which, luckily, does not cause much damage, but which can sometimes have unexpected consequences. Reputedly linked to excessively damp conditions, it has also been known to develop in full sunlight.



Picture 4.29 *Corticium salmonicolor* – Pink Disease. The fungus produces a salmon-pink crusty coating on the surface of the bark which pales later on. From: Nestlé.



Picture 4.30 *Corticium salmonicolor* – Pink Disease. General aspect of a coffee tree severely affected by Pink Disease. From: J. Avelino – CIRAD-CP.

As is the case in cocoa cultivation, different levels of vulnerability to Pink Disease have been detected between the varieties and clones of coffee.

4.2.3.12 Burn or Blight

French	La Brûlure des pointes des rameaux
Spanish	Requemo, Derrite, Quema
Portuguese	Phoma

Causal organism
Phoma costaricensis.

General comments

This disease prevails more specifically at high altitudes in Central America but it can be found throughout the American continent.

The initial symptoms of “Burn” appear in the form of small necrosed spots of an

Pink Disease can be treated with cupric fungicides, but the diseased branches should be removed and burnt.

Tephrosia candida is known to be particularly susceptible to this disease and should, therefore, not be used as temporary shade or as a cover plant in coffee plots.



Picture 4.31 *Phoma costaricensis*. Affected leaves showing necrosed tissues with the typical non-symmetric development into the sickle shape. From: J. Avelino – CIRAD-CP.

undefined shape on the edges of the young limbs of the first pair of leaves. When the attack is of minor importance, the leaves grow in a sickle shape as the limb does not develop at points where the fungus has taken hold. When the attack is more intense, the dark spots spread to cover the entire surface of the leaf. Tiny nubs then appear, these are the pycnids of the pathogen which will develop the spores. The terminal buds are attacked as well as the young shoots which then appear to be totally necrosed. This happens when *Colletotrichum* sp. combine their activity with phoma. This form of the disease is more frequently to be found on young plants or stumped plants and can hinder, or even prevent, the growth of coffee plants. These attacks are often associated with insect damage.

The propagation of the disease is greatly increased by low temperatures, persistent fine rains and winds. Adequate shade and wind-breaks can efficiently prevent it. In the more-exposed zones, fungicide treatments are also recommended. Previously, Captafol was most efficient but since it has been forbidden, cupric fungicides or organic fungicides like Rovral are advised.

4.2.4

Berry Diseases

4.2.4.1 Coffee Berry Disease (CBD)

French	Anthraxnose des baies
Spanish	Antracnosis
Portuguese	Antracnose
German	Kaffeeirschenkrankheit

Causal organism

The fungus which causes this disease is a *Colletotrichum* known for many years as *Colletotrichum coffeanum* Noack as no distinct morphological characteristics could be attributed to it. Later on, to highlight its specific action on berries and to distinguish it from the *Colletotrichum* which attack leaves, this variety of *Colletotrichum* was called “*virulans*” [41] and then, in honor of the extensive studies on the action of *Colletotrichum* on coffee contributed by [25], its name was changed to *Colletotrichum coffeanum sensu* Hindorf. Finally, in order to avoid further confusion and to highlight the specific characteristics and impact of this pathogen, it is now called *Colletotrichum kahawae* Waller and Bridge [55].

General Comments

CBD was first detected and identified by McDonald in Kenya in 1922. It spread rapidly, first to the Kivu district of the Democratic Republic of Congo, and then on to Uganda, Burundi, Ruanda, Tanzania and Angola. In Cameroon, the disease was identified in 1958 by R. A. Muller and, far later, in 1970, in Ethiopia [36]. It only affects Arabica plants and, to date, limited to the African continent.

The impact of the disease on productivity is stronger in high-altitude plantations, above 1500 m in the tropics, but, occasionally, it takes hold in plantations at lower altitudes which have a specifically cold and

damp micro-climate, similar to the prevalent climatic conditions found at higher altitudes.

From the point of view of Arabica cultivation worldwide, CBD would not appear to be a major problem. On the other hand, its impact has significant consequences on the losses. Surveys carried out over a number of years in Cameroon by Muller [35] and Bieysse [8], reveal total losses which can be as high as 80%, or more, of the total production. The intensity of the attacks is closely related to climatic conditions as persistently cool, wet weather favors a rapid development and spread of the fungus. If, at the same time, the berries are plentiful and dense, the fungus propagates even faster and the impact of the disease is disastrous. CBD is a serious threat for all the main growing zones, particularly in Latin America.

Though CBD is still contained on the African continent, the threat of its spreading to other continents is real. South and Central America are particularly vulnerable as Arabica cultivation is widespread and climatic conditions in many areas are similar to those found in the high altitude plantations of Africa.

Symptoms

CBD can be found on all berries at every stage of their development but only the symptoms detected on young green berries can lead to a clear diagnosis of the disease.

On young berries, symptoms appear in two forms:

- “Scabs” which are buff-colored, slightly concave spots of irregular shape with scattered black pinpoints. This form is of minor importance as it only damages the stalk of the berry and does not greatly impact the yield.
- The “active” form of the disease, on the other hand, generates darker wounds which are more concave and carry a greater quantity of acervuli on the injured surface. The pinkish mass of spores develop rapidly. The fungus destroys the pulp and the bean in a few days, leaving little black, empty pouches instead of healthy young berries. The berries are dry and mummified and most of the afflicted berries fall off. The “active form” of the disease causes the losses.

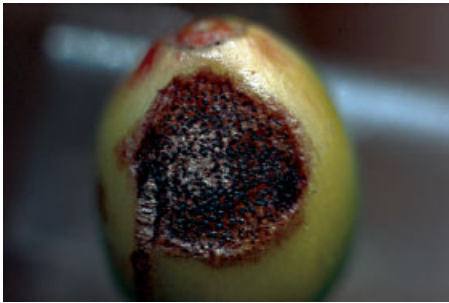
It is interesting to note that the “active” form of the disease chiefly attacks young



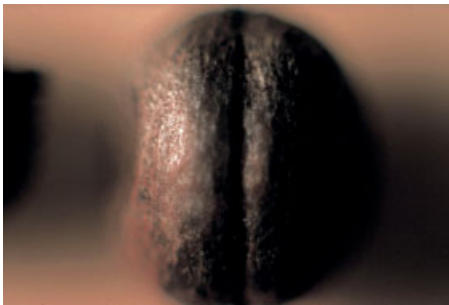
Picture 4.32 *Colletotrichum kahawae* – CBD. Berries affected by CBD. From: A. Autrique.



Picture 4.33 *Colletotrichum kahawae* – CBD. Young Arabica berry affected by the “scab” form of the disease. The concave fawn-colored spots are of irregular shape, slightly corky. A few black specks called acervuli arise on the surface of the spot. The lesion evolves slowly and without gravity. From: R. A. Muller – CIRAD-CP.



Picture 4.34 *Colletotrichum kahawae* – CBD. Young cherries of *C. arabica* affected by the “active” form of the disease. The dark brown concave spots look slimy. They are colonized by many well-defined pinpoints, the acervuli of the pathogen which produce the spores. This damp rot evolves rapidly and affects the pulp of the fruit as well as the seeds. From: R. A. Muller – CIRAD-CP.



Picture 4.35 *Colletotrichum kahawae* – CBD. Close-up of a young Arabica berry affected by the “active” form of the disease which has reached the final stage of decay. The fruit is completely destroyed. From: R. A. Muller – CIRAD-CP.

berries, whereas the “scab” form is found on more mature berries. This would seem to indicate that the “scabs” are, in reality, generated by a resistance of the tissues to the disease.

CBD symptoms which appear on mature berries are not characteristic of the disease. Known as “Brown Blight”, they closely resemble the attacks of other saprophytes (*Colletotrichum*) and extend to the pulp of the berry to cause premature ripening or

softening. They have very little impact on the crop itself as the coffee beans are rarely damaged and only occasionally touched by the blight.

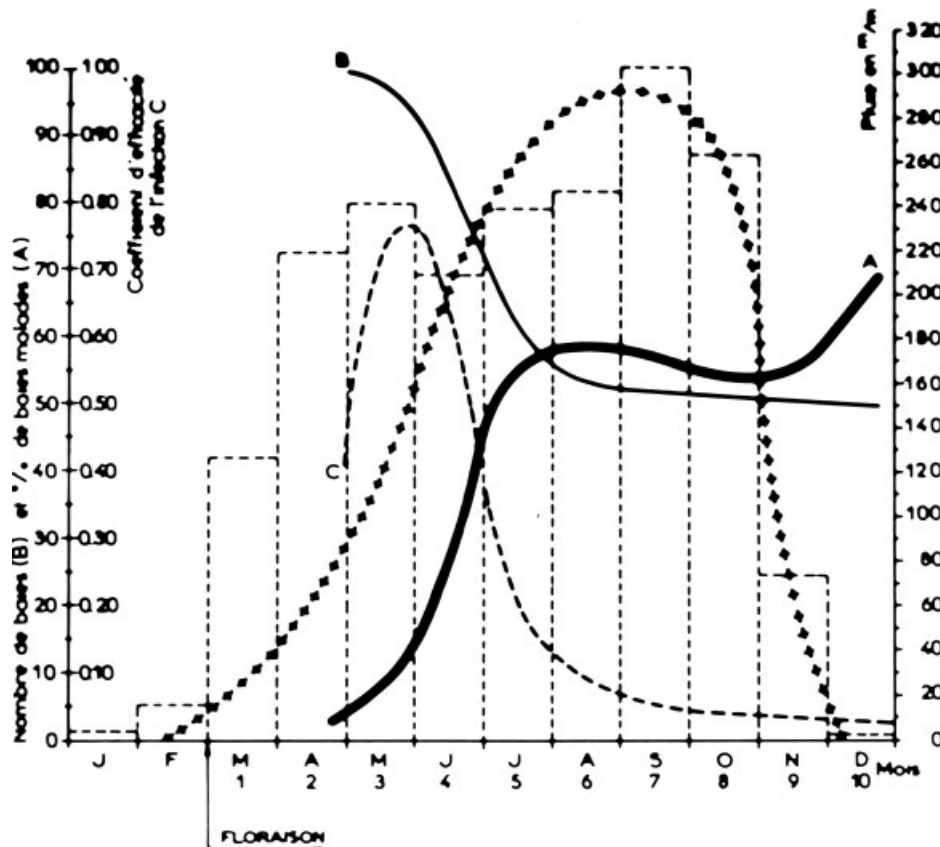
To complete the diagnosis and in order to avoid confusion, other significant aspects should be highlighted:

- CBD attacks very green berries on coffee trees which are otherwise healthy and have a well-developed vegetative growth. It should not be confused with attacks of *Cercospora coffeicola* which attack the leaves and berries of trees suffering from a lack of nitrogen. These trees are already yellowed and the symptoms caused by the pathogen have a different aspect.
- CBD can destroy practically all the berries on the coffee tree without having any impact on the leaves or the branches. This point is very significant when it comes to establishing a correct diagnosis of the afflicted coffee trees and clearly distinguishes the pathogen which attacks berries from other pathogens which attack leaves, branches or the entire tree.
- Berries destroyed by CBD are in no way to be confused with the dried, blackened pouches left hanging on an equally withered and blackened branch after an attack of die-back.

Propagation

The great number of spores of the pathogen generated on the diseased berries are dispersed by rains and transported from tree to tree by anything that moves through the rows of the plantation. Wind, however, is not a major propagation factor. The development of the disease to a level of epidemic importance depends chiefly on rains and fog as the spores of the pathogen need moisture to germinate and develop.

Studies carried out in 1969 by S. K. Mulinge in Kenya and, more importantly,



Picture 4.36 *Colletotrichum kahawae* – CBD. Diagram showing the coefficient of intensity of the infection (C) comparing the number of berries lost (B) and the percentage of infected cherries

(A) against the rainfall (in mm) over the months of the year. Blossoming is given to be at the end of February. From: R. A. Muller – CIRAD-CP.

by R. A. Muller in Cameroon show that the susceptibility of the berries varies during the different stages of their development.

R. A. Muller has drawn the following conclusions from the chart (see Pict. 36):

- 1) The evolution of the disease is expressed by the percentage of infected berries in ratio to their total number. This curve (*thick line*), based on weekly observations, follows the stages of the development of the berries and indicates periods of expansion, stabilization,

regression and a renewed outbreak in the evolution of the disease. Where the period of expansion can easily be explained by heavy rains which favor the development of the pathogen, the periods of stabilization and regression, which take place in the same climatic conditions, can only be attributed to different factors which do not depend on climatic conditions. The same would apply to the renewed outbreak of the disease, noted at the beginning of the dry season (October–November), which

would not seem to be conducive to the germination and dissemination of the spores.

- 2) This curve (*thin line*) shows the losses due to CBD with a peak at the end of April, 2 months after blossoming, followed by a regression and a period of little variation.
- 3) This curve (*dotted line*) shows the evolution of the coefficient of “susceptibility” of the berries in both quality and quantity. It rises rapidly to a sharp peak when the berries are young and quickly stabilizes when the berries are in their fifth month of development. This coefficient is calculated on a weekly basis taking into account the number of berries lost over 2 weeks “ $n + 1$ ” and “ n ”, and the number of berries existing in week “ n ”. This susceptibility coefficient is independent of climatic conditions. It depends on the physiological stage of the development of the berries. This could possibly explain the periods of stabilization and regression as well as the new outbreak of the disease shown in curve (A). The periods of stabilization and regression could be attributed to the fact that the development of the berries had reached a stage of “non-susceptibility” during their maturation (July), while the new outbreak of the disease (November) could be caused by the fact that once the pulp had ripened, its

sweet, juicy content offers the favorable conditions needed for development of the fungus remaining in the berries after the initial attack of the pathogen. In this case, the new outbreak of the disease would not mean a new infection of the pathogen but rather a regeneration of the initial infection. Table 4.5 sustains this hypothesis.

Further observation also shows that, during the period of “non-susceptibility” of the pulp, the parchment hardens and thickens, providing a strong barrier against the invasive pathogen and efficiently protecting the coffee bean.

Hence the above observations would lead to the hypothesis that the disastrous impact of the disease is due to the fact that heavy seasonal rains, which favor the development of the spores of the pathogen, coincide with the stage in the development of the berries where their susceptibility is at its peak.

As mentioned previously, the pathogen is common in coffee plantations at altitudes above 1500 m close to the Equator. The pathogen has, however, been observed occasionally at lower altitudes under microclimates similar to the high altitude and specially in plantations which cultivate Caturra, varieties known to be especially exposed to the pathogen. In Cameroon, observations made on Caturra have revealed

Table 4.5 Moisture content of fresh pulp and beans, shown as % of their weight

	Age of the berries (weeks)										
	8	11	13	15	17	20	22	26	28	33	35
Pulp (%)	80.2	83.3	86.5	78.4	77.7	74.8	72.9	72.3	72.5	79.8	77.4
Beans (%)	88.6	91.0	91.5	90.7	90.8	88.5	86.8	74.2	69.7	53.3	43.9

From: G. Blaha – CIRAD-CP.

that vulnerability is marked in plantations situated at an altitude of 1100 m, whereas *C. arabica* var. *Jamaica* is severely afflicted at an altitude of 1500 m or more, but not at 1100 m. Studies carried out by R. A. Muller in 1980 have led to the conclusion that this phenomenon is due to the fact that, at low altitudes, the berries develop much faster, and thus the pathogen has less time to take hold and penetrate them. At a higher altitude, however, growth is correspondingly slower and offers the pathogen more time to infect the berries. This theory is substantiated by the fact that maturation cycle of the same variety of Arabica takes 32 weeks at 1100 m and 42 weeks at 1500 m. This, however, remains a theory which requires further research to justify changes in cultivation practices.

Chemical control

In geographical zones like Cameroun, Malawi or Zimbabwe which only have one rainy season and one flowering period, the spray program is based on the evolution of the disease given earlier in this chapter with curves highlighting the peaks and regression of the impact of the pathogen that clearly indicates the best periods for scheduling chemical treatments. The spray program should begin just after the blossoming of the coffee trees to protect the young berries before they reach their first stage of "susceptibility" and should be repeated at precise intervals calculated according to the remanence of the chemical used. The frequency of the treatments could, however, be increased in the case of important rainfall. Treatments should stop at week 20 after blossoming, at which point the berries are no longer vulnerable to the pathogen.

Thus the basic schedule for cupric fungicides would be as follows:

Treatment 1	just after flowering
Treatment 2	5 weeks after the first treatment
Treatment 3	4 weeks after the second treatment
Treatment 4	3 weeks after the third treatment
Treatment 5	2 weeks after the fourth treatment
Treatment 6	2 weeks after the fifth treatment
Treatment 7	2 weeks after the sixth treatment

In Cameroon, the recommended fungicides are NORDOX 50 WP (cuprous oxide 58 %), NORDOX SUPER 75 WP (cuprous oxide 86 %) and KOCIDE 101 (cupric hydroxide 56 %).

Over recent years, several molecules have been developed and tested with a view to broadening the scope of chemical treatments available, and, more importantly, to generate efficient systemic, in-depth treatments which would reduce the number of applications required to protect the plantation against the pathogen.

It should be noted that in tropical climates, like in Cameroon, where there is only one flowering during the year, seven applications of cupric fungicides are required to efficiently protect coffee trees.

In equatorial bimodal climates like Kenya, however, where there are two annual flowering periods, berries generated by the April flowering contaminate those generated by the October flowering. Similarly, berries developing from October onwards would contaminate those developing in April. In this way, on plantations in bimodal climates, contamination is ongoing throughout the year so that 10 or 12 annual treatments are required. This, inevitably, means that high doses of cupric fungicides are used extensively. While copper, in small

doses, has a tonic effect on coffee trees, the massive doses required to provide efficient protection against CBD have a toxic effect, sapping their vitality and impairing their vegetative growth. Nevertheless, and in spite of these drawbacks, cupric fungicides are still used extensively, although, more often, they are now mixed with efficient, but more expensive, organic pesticides. Mixing them with cupric fungicides reduces costs and offers the added advantage of immediately destroying any strains of the pathogen that develop a resistance to organic pesticides. In Kenya, in certain areas where *Pseudomonas syringae* pv. *garcae* is active at the same time as CBD, the use of cupric fungicides remains essential (Tab. 4.6). In this case, cupric hydroxide or “Bordeaux Mixture” is recommended.

In countries where the climatic conditions only favor one major annual blossom-

ing, the situation is somewhat different, but there are significant similarities. For instance, berries generated by small intermediary flowerings are not picked during the main harvest, but they are contaminated and could, therefore, transmit the pathogen to the next crop. The same applies to leftover berries on the trees by the pickers, those that have dropped to the ground as well as the peduncles which remain on the tree.

Based on the assumption that the berry-to-berry contamination is a major factor in the propagation of the disease and that the denser the berries, the greater the contamination, it becomes obvious that the epidemic will be far more serious when the coffee trees produce a high yield. Consequently, it is recommended to specifically schedule chemical treatments on the basis of the importance of the crop and

Table 4.6 Tested and recommended fungicides (active substance) for CBD control and tank mixture for CBD and CLR control in Kenya

CBD control Active substance	CBD and leaf control (tank mixture)	
	Organic fungicide	Copper 50% WP
Chlorothalonil	Daconil 2787 W75	cuprous oxide
Fluazinam		cupric chloride
Dithianon		cupric hydroxide
Azoxystrobin	Delan 75 % WP	cuprous oxide
Anilazine		cupric chloride
Anilazine/copper		cupric hydroxide
Chlorothalonil/copper	Dyrene 75 % WP	cuprous oxide
50 % Copper formulation		cupric chloride
(a) Cuprous oxide		cupric chloride
(b) Cupric chloride	Rova 75 % WP	cupric chloride
(c) Cupric hydroxide		
Copper sulfate Lime		
	Quadris	cupric chloride
	Delan 500 SC	cupric chloride

From: Technical Circular 804-2003.

the density of the berries on the branches. Dwarf varieties with short internodes are more vulnerable than taller trees with longer internodes.

As to the methods used for applying chemical treatment, both past and recent research lead to the same conclusion: high volume spraying of 700–1000 l of brew/ha are the most efficient and spraying should aim at the tops of the trees. In Kenya, special, mechanical pulverizer have been designed for this purpose; they “shower” the trees from the top downwards. Pneumatic low-volume spraying is generally less efficient, but is widely used in Ethiopia where the height of the trees entails the use of powerful mechanical equipment to reach the canopy which is not accessible with current knapsack sprayers.

On the other hand, supposedly efficient results have been obtained by ultra-low-volume spraying techniques in Kenya. These techniques are tempting because they do not consume a lot of water, but it is not certain that they are really adapted to fighting fungus as the most efficient fungicides need to be sprayed in large volumes throughout the coffee tree. Ultra-low-volume techniques could, however, provide better results with new organic pesticides whose action is not geared to protective covering, but to penetration within the plant at specific points. Once experiments with this type of treatment have proved its real efficiency and implementation has become current, significant changes will take place in the fumigation practices used for chemical control. Meanwhile, as discussed earlier in this chapter, simple “impluviums” can be cheaply and easily installed on the plantation to provide the water necessary for high-volume applications on the coffee trees.

Picture 4.34 shows the differences between a crop that had been efficiently pro-



Picture 4.37 *Colletotrichum kahawae* – CBD. Aspect of a coffee crop treated against CBD with cupric compounds (plot 3) as compared to a non-treated control (plot 2). From: R. A. Muller – CIRAD-CP.

tected from the pathogen by cupric compounds applied at high volume as compared to that of a control tree which had not been treated.

Agronomical control

Based on the evolution of the “susceptibility” cycle of coffee fruits during their development, R. A. Muller set up an “early irrigation strategy” in Cameroon. Investigated, experimented and implemented

over a period of 10 years between 1973 and 1982, this strategy has been proved extremely valid and provides most satisfactory results.

The strategy consists of providing adequate irrigation to the coffee tree as early as the middle of January; in other words, artificially generating climatic conditions which are the same as those which prevail at the onset of the rainy season. This induces the dormant buds on the tree to flower during the dry season, approximately 1.5 months before their normal blossoming at a time which does not favor the activity of the pathogen.

Once the rainy season sets in and the pathogen becomes active, the berries will have already outgrown their “susceptibility” stage. Thus, the majority of them will not be infected by the pathogen.

Early irrigation strategies, combined with a distinctly reduced number of chemical treatments, have also been noted to increase the productivity of Arabica coffee trees. Consequently, such cultivation practices have been introduced in Cameroon with highly satisfactory results on plantations where sufficient water was available during the dry season to ensure adequate irrigation. Large-scale implementation would, however, require installing an adequate irrigation system on all the plantations threatened by CBD.

This would, undoubtedly, provide an efficient method for controlling a lethal disease which combines environmentally friendly cultivation practices with limited chemical control and offer the added advantage of increasing both the quality and the quantity of the coffee.

Integrated control

The convincing results obtained by implementing early irrigation strategies would appear to be an important factor when

examining the different aspects of “integrated cultivation practices” or the ways and means of combining all available strategies to improve disease control and the prosperity of the plantation as a whole. To date, however, little research has been carried out to further develop other practices and only a few useful strategies can be recommended.

The following points should, nevertheless, be considered:

- 1) In regions where only one major flowering takes place during the year, special care should be taken, after harvesting the main crop, to remove all remaining berries, even those that are not ripe. These berries are carriers of the pathogen and would, therefore, contaminate the following crop. Removing them delays the impact of the disease, a fact which has been substantiated experimentally by R. A. Muller in Cameroon. Furthermore, it is also a preventive measure against the Coffee Berry Borer *Stephanoderes hampei*.
- 2) It would also appear that an excess of potassium chloride generates a situation which is favorable to the CBD pathogen. The same applies to zinc sulfate or zinc chelate.
- 3) Similarly, certain fungicides, particularly cupric fungicides, also seem to favor the sporulation of the CBD pathogen as well as an increase in the activity of other related pathogens. This theory, developed by Gibbs in 1972, has, however, never yet been fully researched and substantiated, and cannot, therefore, lead to any concrete recommendations. It is suspected that fungicides destroy the antagonist microflora and thus create conditions which favor epidemic development [31].

- 4) Recent observations in Cameroon show the incidence of the agro-environmental situation of the coffee plantation on the development of the epidemic. Shading, for instance, has a negative impact on disease development. These studies also revealed the importance of “physiological berry fall” in relation to cultivation conditions. Precise knowledge of climatic conditions favoring disease development and of the phenological cycle of the trees would help advisors to make recommendations adapted to different types of farms, in order to implement optimal management strategies and decrease phytosanitary treatments [9].

Genetic Control

Researching resistant varieties, another aspect of integrated farming practices, has been ongoing for a number of decades [21].

Initiated by Cook [15], and carried on by van der Vossen [54], research was carried out with a view to developing specific oligogenic resistance based on wild varieties like the Timor Hybrid and Rume Sudan, which both showed diverse stages of resistance characteristics. Experiments relied on systematic tests carried out by inoculating young trees at the “soldier” stage whose reactions were judged to be similar to those of the berries. As a result of this experimentation, the Catimor variety, which also shows valid resistance to CLR, was defined as the most interesting possibility. Variety Ruiru 11 appears to a worthwhile choice as it combines resistant strains to both *C. kahawae* and to *H. vastatrix*, but, again, suitable precautions need to be taken as certain Catimor descendants already show signs of vulnerability to the pathogen.

In Cameroon, experiments carried out by Bouharmont [11] led selection of the Java variety as offering promising resistance to CBD and CLR as well as the added advan-

tage of being vigorous and high-yielding with an excellent cup quality. The resistance level of the Java variety has, however, yet to be put to the test as, for the time being, this does not appear to be a constant characteristic. Further selection to better define it would appear to be necessary.

When CBD were first detected in Ethiopia in 1970, the center of the diversity of *C. arabica*, a broad scope of possibilities for research on a number of Arabica varieties was opened to researchers like Robinson [44] and van der Graaf [53]. It became possible to observe the reactions of a great number of varieties of *C. arabica* and to determine those which offer the most promising resistance to CBD.

Once this was achieved, it became possible to select specific varieties to be used for experimentation and research prior to introducing them into plantations as suitably resistant varieties on a long-term basis.

More recently, a project financed by the European Union has grouped the efforts and results obtained by CIRAD in France, CIFC in Portugal, IRAD in Cameroon and CRF in Kenya. This has greatly developed research into genetic resistance to the CBD pathogen.

In its first stages, this international co-operation program identified two distinct groups of strains: the East African strains and the Cameroon strain, as well as several bridge strains. This identification was based on research carried out on a great number of strains of the pathogen collected in various African countries where CBD onslaughts were rife. The geographic and genetic diversity of *C. kahawae* was then drawn up by Bella Manga *et al.* in 1997. Molecular markers, using randomly amplified polymorphic DNA (RAPD) analysis techniques, have led to the conclusion that there is not, however, a very significant genetic variability between the two groups.

At the present stage of this research program, it has not yet been possible to define the type of resistance coffee generate against *C. kahawae*, the CBD pathogen. This seems to lead to the conclusion that we are faced with a “general” as opposed to a “specific” type of resistance. It has also been possible to identify different strains which show different characteristics of the disease within the pathogen itself. Thus, CBD strains in Cameroon appear to show more aggressive characteristics than those in Eastern Africa. These findings point to the possible necessity that selection made in certain geographical zones will have to be repeated on local pathogen strains.

Apart from the study of the genetic structure of *C. kahawae* populations, the program also includes the study of the development processes of the infection itself. This involves, among other things, an evaluation of environmental influences, a histopathological study in the field of the reactions of the berries of different genotypes to the pathogen, and the setting up of precise and reliable methods of observation to assess losses caused by the disease.

The major target this program has been set is, however, to determine the best sources of resistance to the pathogen and the corresponding mechanisms of the transmission of this resistance in order to better assess the resistance of new descendants issued from the breeding of identified genotypes.

To achieve this result, a reliable system of testing is required. At face value, it would seem obvious that tests should be carried out on the berries as they are the vulnerable organs of the coffee trees. This, however, is a time-consuming procedure as it is necessary to test each annual fructification over a number of years before obtaining reliable results.

To save time, Kenyan research, as already mentioned, was carried out in research nurseries on young trees at the “soldier” stage, believed to replicate the reactions of the berries in the field. As no other testing method has yet been investigated, this method is still implemented today, but as it does not involve the actual berries in the field, it cannot be considered truly reliable until further proof is established. On the other hand, it is quite likely a useful tool for the first stages of selection as it would indicate potentially exploitable new genotypes, prior to their actual field test.

Thus, active research work is now ongoing in the hope that a genotype with true resistance to the CBD pathogen will be developed soon.

Assessment of losses caused by CBD

Global assessment of losses caused by CBD on any given plantation is relatively easy as the final yield of the annual crop provides an indisputable answer. While this may be adequate when it comes to an overall evaluation of the efficiency of chemical treatments, it is totally inadequate when precise studies of the epidemiology of the disease are required in a given area. Thus, a “physiological berry fall” can reach 50% of the potential crop and can be confused with CBD losses.

In Cameroon, weekly berry counting has provided a useful tool as it also distinguishes between CBD diseased berries and “physiological berry fall” with “scabs”. This method has been improved by not only counting the berries on marked branches where the number of berries had previously been established, but also by delicately marking diseased berries, detected during a given week count, with a label attached to a fine thread. In this way, the next weekly observation can also distinguish the number of “physiological

berry fall” from the number of berries lost because of the disease. This method is, obviously, extremely painstaking, but, given the rapid rate at which young berries drop at the vulnerable stage of their development, it seems, to date, to be the only way to clearly define the impact of the disease. These few comments on the considerable difficulties encountered by researchers when investigating the impact of the disease are only mentioned as an indication of the precision and care taken to either substantiate or invalidate the various theories and questions that arise in the course of a scientifically conducted research program.

Conclusion

The many different methods which can be implemented to fight CBD, chemical control linked to improved cultivation practices and genetic control, among others, indicate that the solution to the serious problem of controlling CBD in Arabica can be considered as an excellent illustration of how integrated cultivation practices based on scientifically substantiated counseling by qualified agronomists can produce the most satisfactory results.

4.3

Bacterial Diseases

4.3.1

Elgon Die-back or Bacterial Blight

Causal organism

Pseudomonas syringae pv. *garcae*.

General comments

This disease was first observed in Kenya, in Uganda and on the slopes of Mount Elgon, above an altitude of 1800 m. In Brazil, it was identified at Garça (Sao Paulo), hence its name “garcea”. In 1945, Thorold, on

the basis of the symptoms of the disease, deduced that it must exist in Zaire or in India, but according to J. Avelino, Elgon die-back is latent in most coffee producing countries, but only develops under very specific conditions.

Symptoms

Basically, Elgon die-back is a nursery disease which essentially affects leaves and young tissues.

When the leaves are damaged, they first appear to be water-soaked. This is the initial stage of the infection. Later on, the leaves curl up, dry out, blacken and die, but they do not drop off the tree. The initial attack affects the terminal buds and works downwards, causing die-back of the twig. A small cluster of blackened leaves generally remains at the tip of the branch. This is the factor which differentiates Elgon die-back from die-back caused by overbearing when the twigs die after losing their leaves.

On young, succulent shoots, infection can be noticed on the stipules of the nodes just below the tips then, progressively, the nodes are totally invaded. Necrosis of the upper part of the shoot is caused by the destruction of the vascular tissues. A typical symptom is the blackening of the part which corresponds to the petiole at the base of the shoot.

Where, occasionally, it affects established plantations, the symptoms appear on flowers and pin-head berries which blacken and fall.

Attacks can affect the entire plant or only parts of it. In severe cases, the coffee tree looks as if it has been scorched by fire (Fireblight).

Biology

The bacteria live in epiphytes on all parts of the plants and remain “expectantly inactive” until the ideal conditions prevail.



Picture 4.38 Elgon die-back on a young shoot.
From: A. Autrique.

When this happens they will become active pathogens. Ideal conditions for *P. syringae* are rain or dampness and cold winds, the type of conditions which are frequent on the slopes of Mount Elgon, the Solai Valley, Paranà or Sur de Minas. According to Kairu, on the basis of observations recorded in 1983, the incubation period of the disease is approximately 4 weeks.

Though high altitudes generate favorable conditions for the development of both CBD and Bacterial Blight, these two diseases are rarely to be found in the same place at the same time.

Control

In 1945, Thorold noted that “French Mission” coffee trees resisted well to Bacterial Blight, and that they were “bronze-tipped” trees with broad and rather coarse leaves.

Further observations led him to record that nearly all the “green-tipped” trees were susceptible to the disease.

If nurseries are well protected against wind with efficient windbreaks, outbreaks of Bacterial Blight are less frequent. If there is potential danger of an outbreak, trimming and pruning should be reduced.

Copper oxychloride is the best chemical control to date. In nurseries it should be applied twice at an interval of 20 days in a solution of 50% Cu at 0.2–0.3% + 0.2% of either Agrimicine or Distreptine 20.

On established plantations, in the field, one or two copper oxychloride treatments with 50% Cu at 1% are recommended.

4.4

Physiological Diseases

4.4.1

Hot and Cold Disease

This affection, also known as Black Tip, was first reported in Uganda. Initially, it was thought to be caused by a virus as its various symptoms can also be typical of viral diseases.

The tips of the young shoots blacken, and the adjoining leaves crinkle, distort and reduce in size. Terminal growth is excessively branched and bunched. The discoloration of the leaves starts from the outer edge and works inwards. In severe cases, the leaves may scorch and fall or, if they are subsequently infected by *Phoma* spp., die-back of the young shoots may follow.

This physiological disease is caused by environmental stress, particularly high temperature variation between night and day. It can also be caused by insect stings, but it only prevails in high-altitude planta-

tions. According to a survey carried out in Cameroon by Bruneau de Mire in the late 1960s and early 1970s, hymenopterous insects (*Ulopidea*, *Collborhis* sp.), whose sting could cause this syndrome, would only be present under specific climatic conditions.

4.5 Bibliography



Picture 4.39 Hot and Cold Disease.

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